



(12) **United States Patent**
Kim

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(54) **SEMICONDUCTOR DEVICE AND METHOD FOR FABRICATING THE SAME**

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(30) **Foreign Application Priority Data**

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(51) **Int. Cl.**
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H01L 21/02 (2006.01)
H01L 21/302 (2006.01)
H10D 30/67 (2025.01)

(52) **U.S. Cl.**
CPC .. **H01L 21/76885** (2013.01); **H01L 21/02225** (2013.01); **H01L 21/02425** (2013.01); **H01L 21/302** (2013.01); **H01L 21/7688** (2013.01); **H10D 30/6728** (2025.01)

(58) **Field of Classification Search**
CPC H10B 12/30; H10B 12/0335; H10B 12/05; H10B 12/09; H10B 12/48; H10B 12/50; H10D 30/6728; H01L 21/76885; H01L 21/0225; H01L 21/302

See application file for complete search history.

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(57) **ABSTRACT**

A semiconductor device may include a substrate; a plurality of semiconductor pillars disposed over the substrate and arranged in a first direction and a second direction crossing the first direction; an insulating layer pattern disposed between the substrate and the semiconductor pillars and extending in the second direction; a first conductive line disposed between the insulating layer pattern and the semiconductor pillars and extending in the second direction; a second conductive line formed over sidewalls of the semiconductor pillars and extending in the first direction; and a storage node disposed over each of the semiconductor pillars.

14 Claims, 33 Drawing Sheets

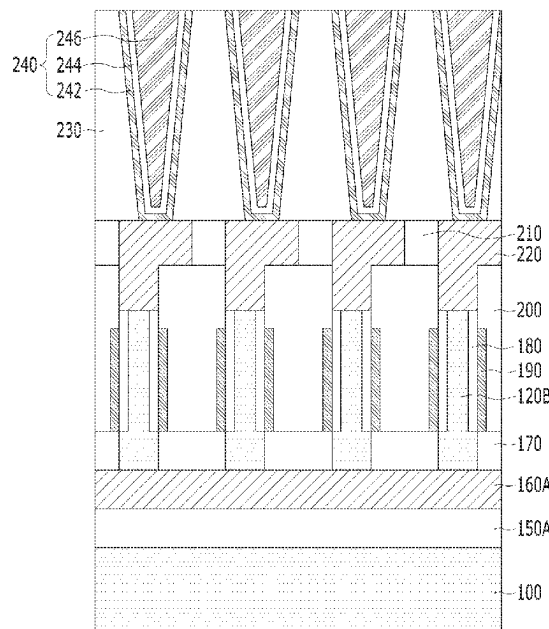


FIG. 1A

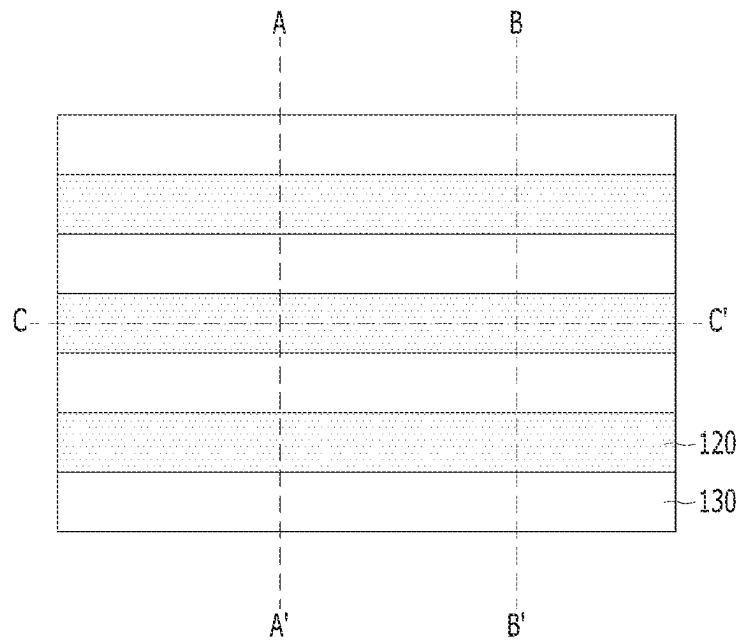


FIG. 1B

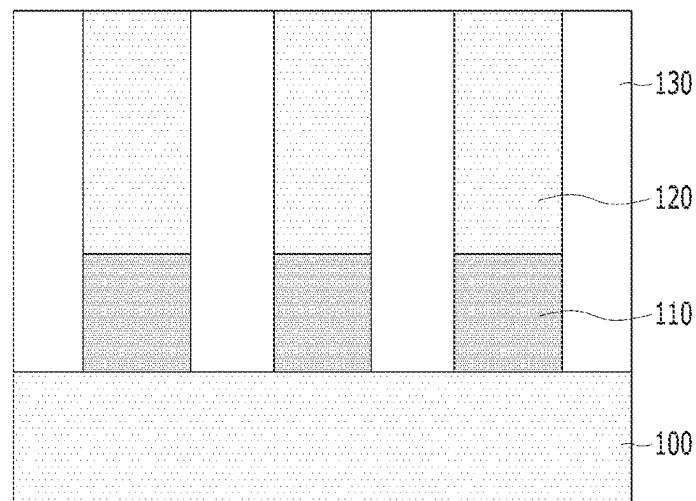


FIG. 1C

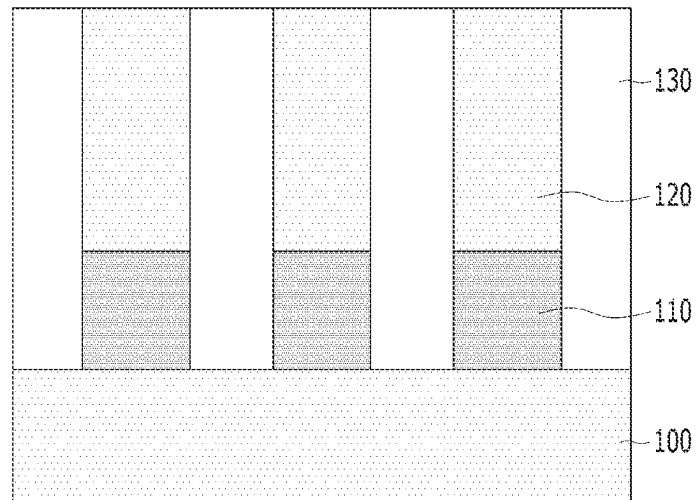


FIG. 1D

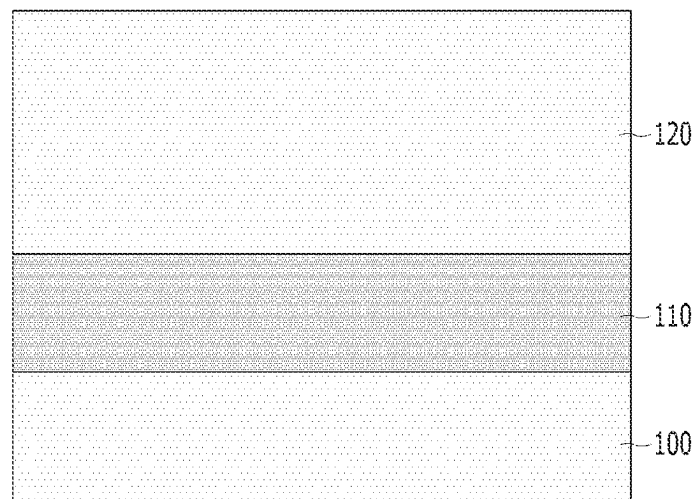


FIG. 2A

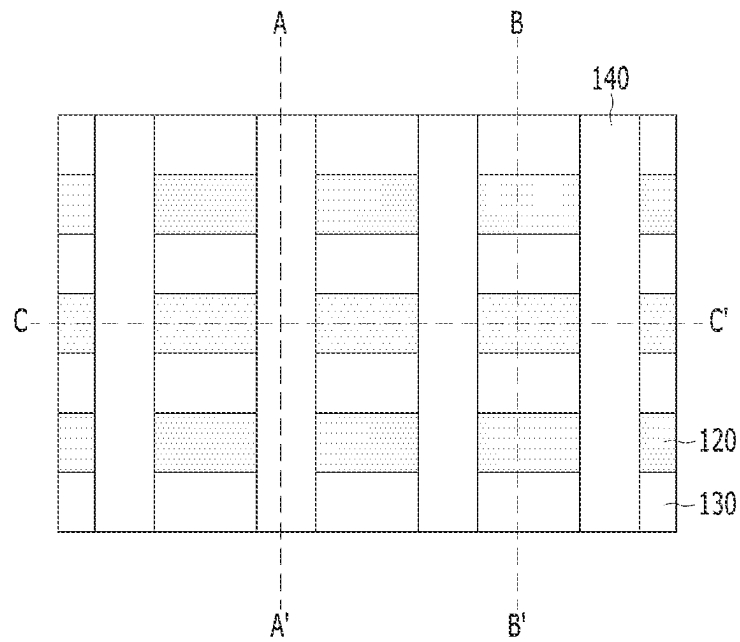


FIG. 2B

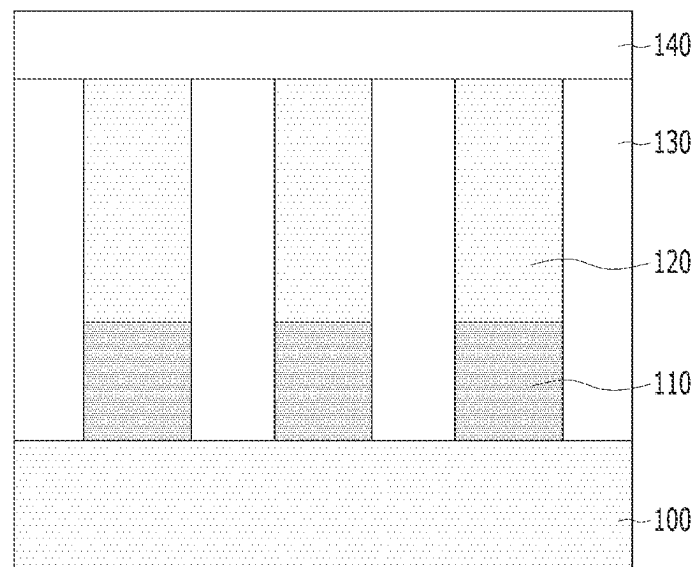


FIG. 2C

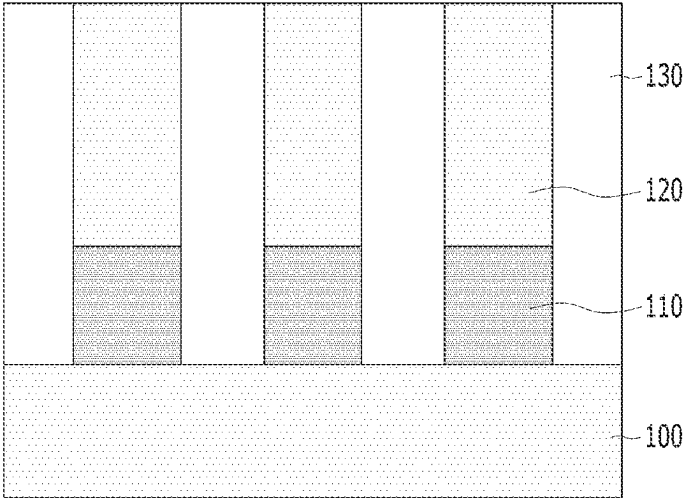


FIG. 2D

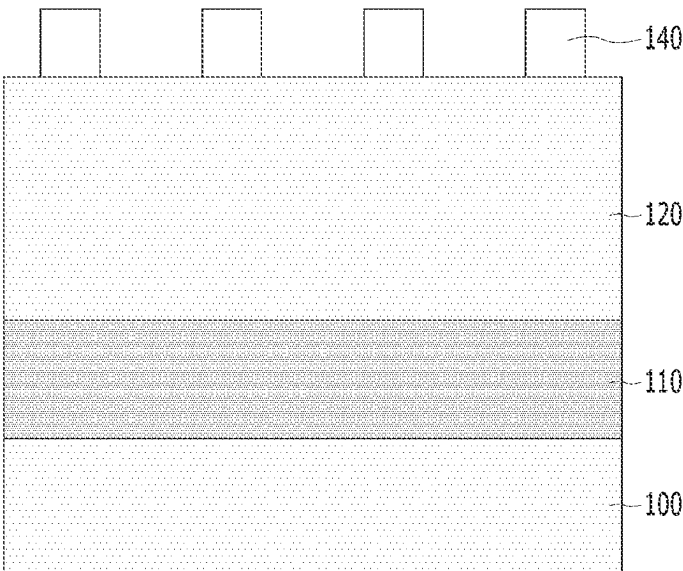


FIG. 3A

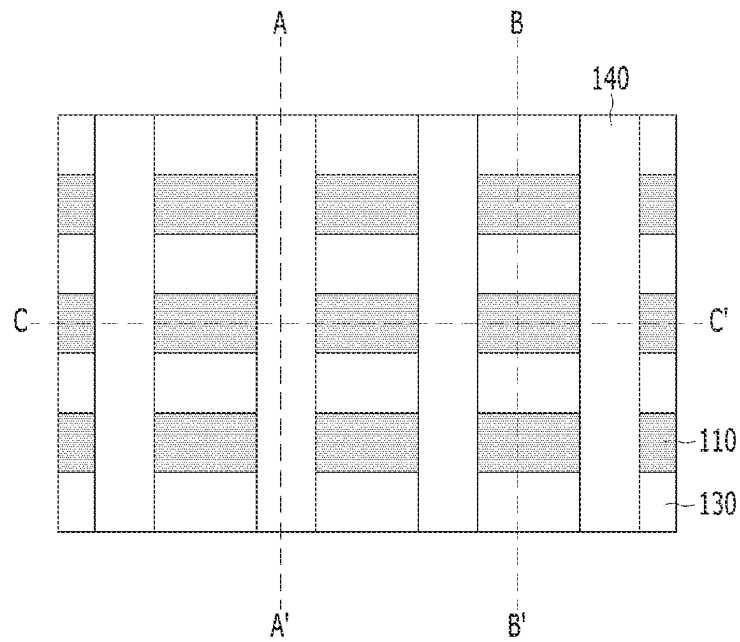


FIG. 3B

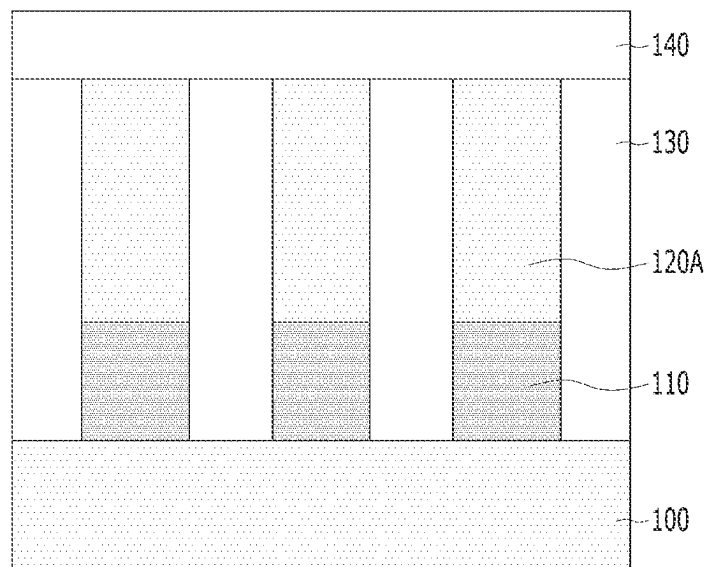


FIG. 3C

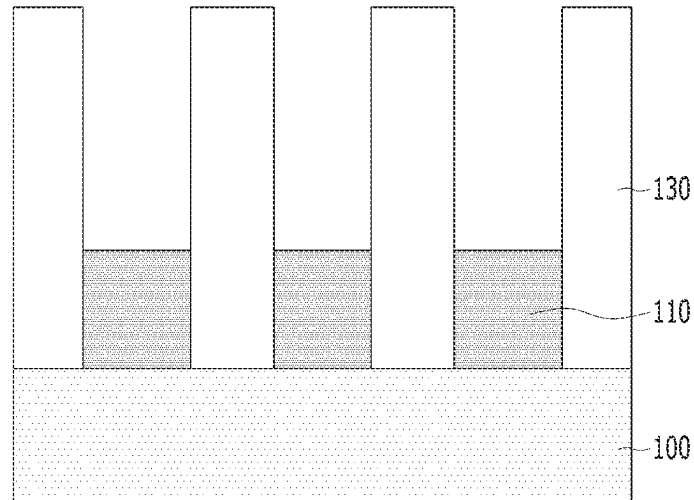


FIG. 3D

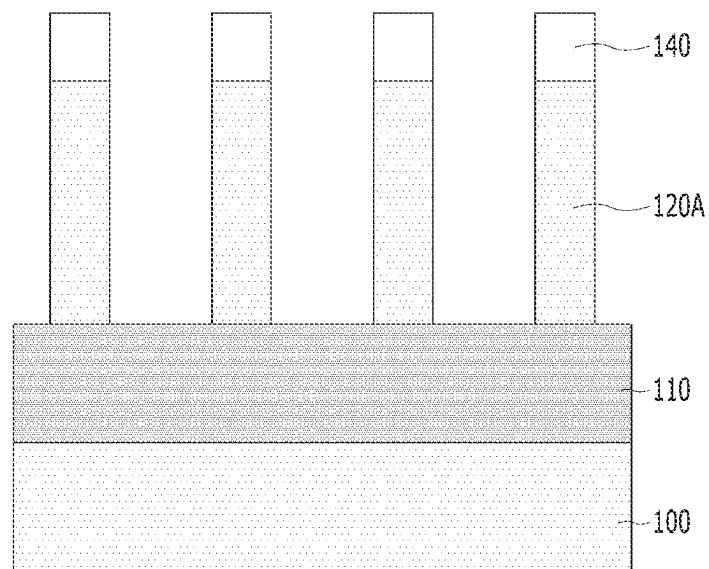


FIG. 4A

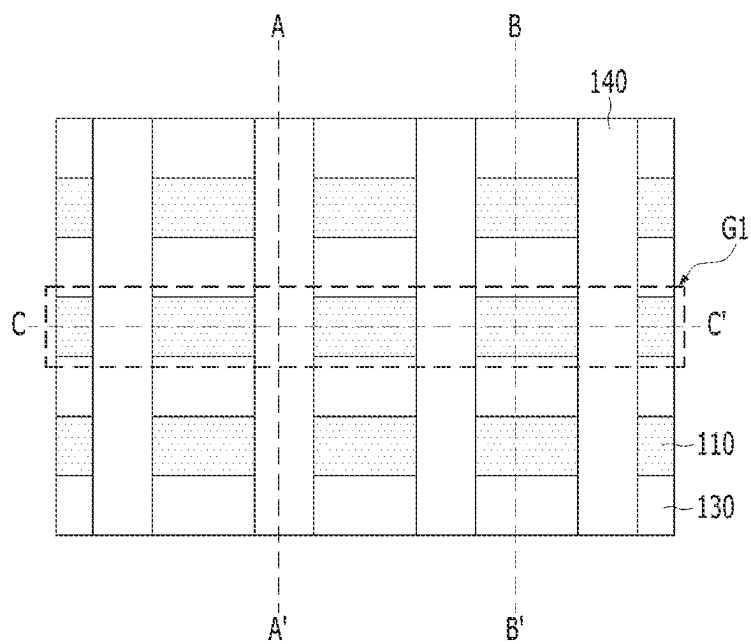


FIG. 4B

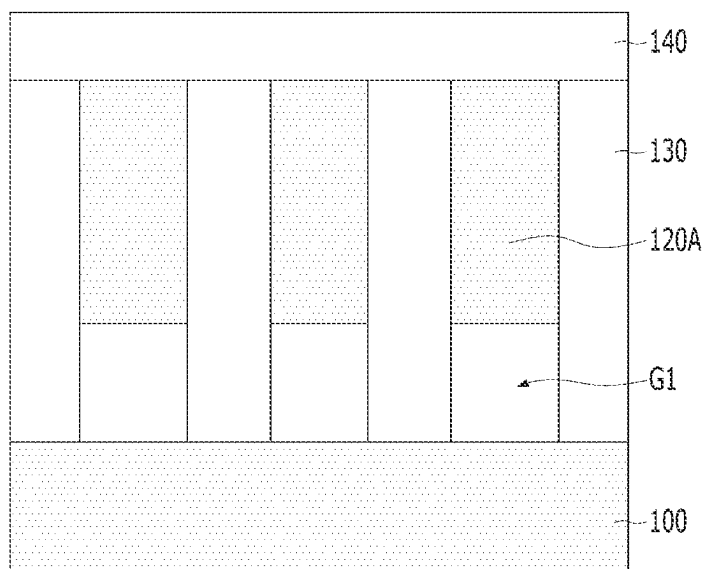


FIG. 4C

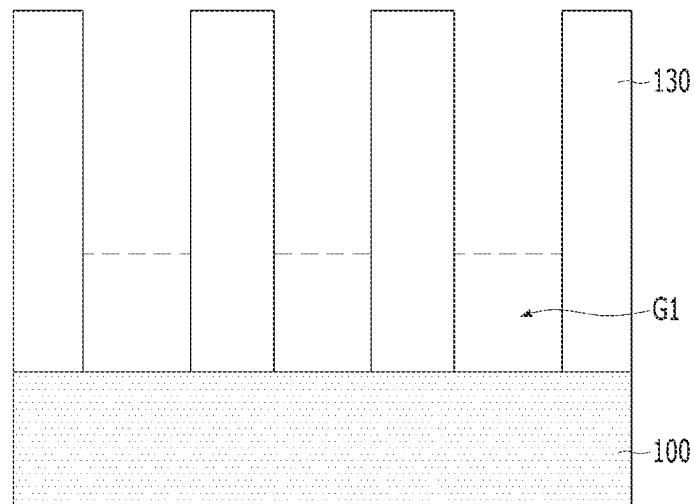


FIG. 4D

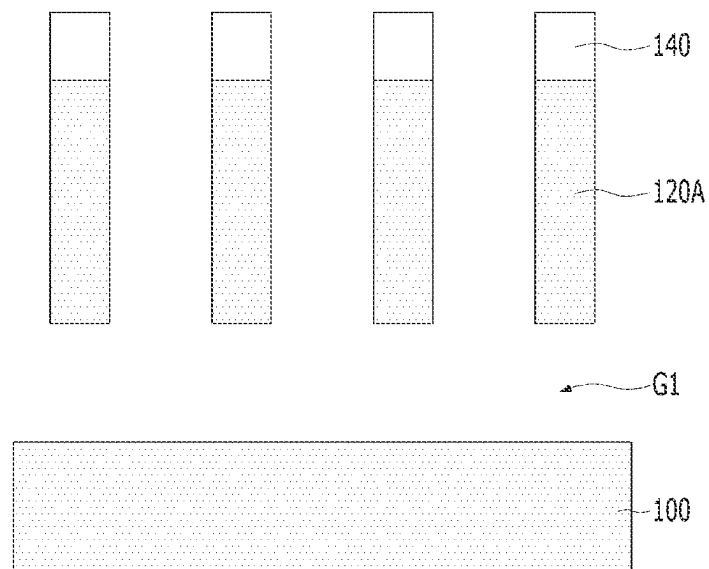


FIG. 5A

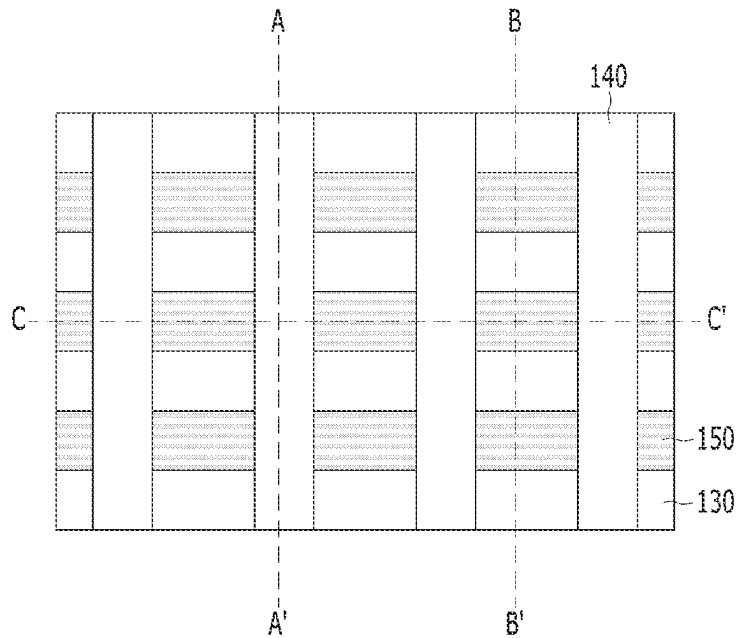


FIG. 5B

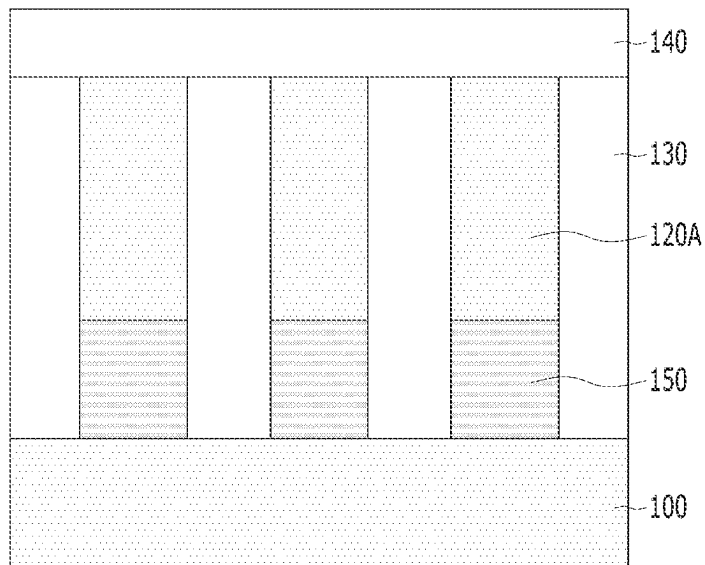


FIG. 5C

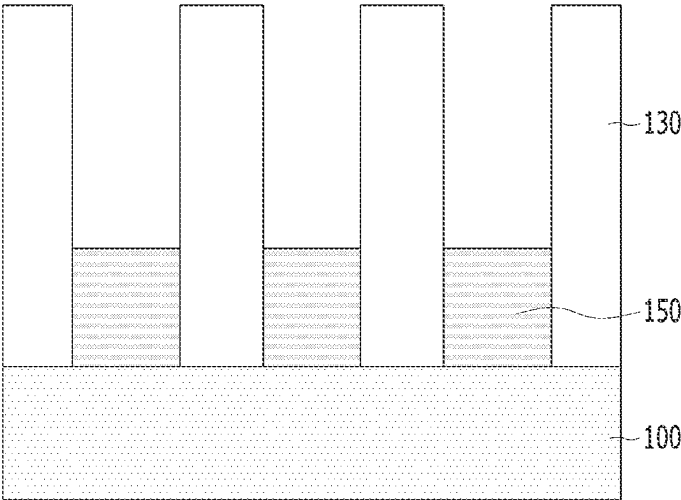


FIG. 5D

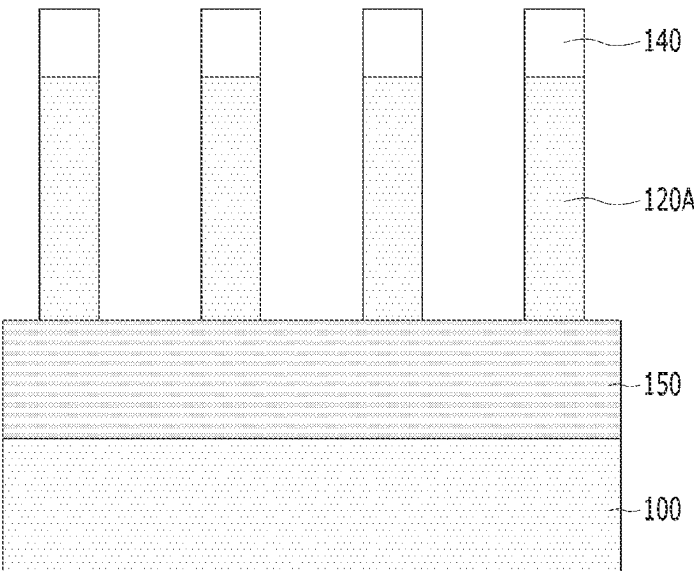


FIG. 6A

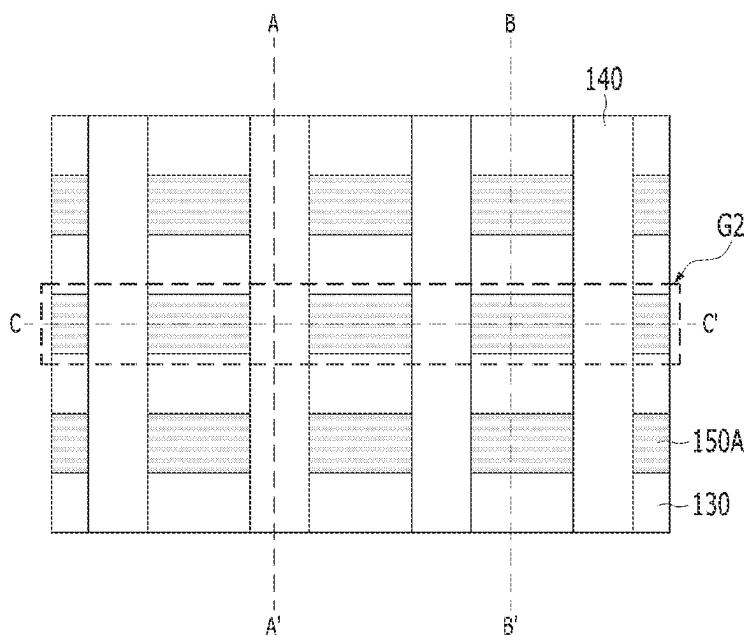


FIG. 6B

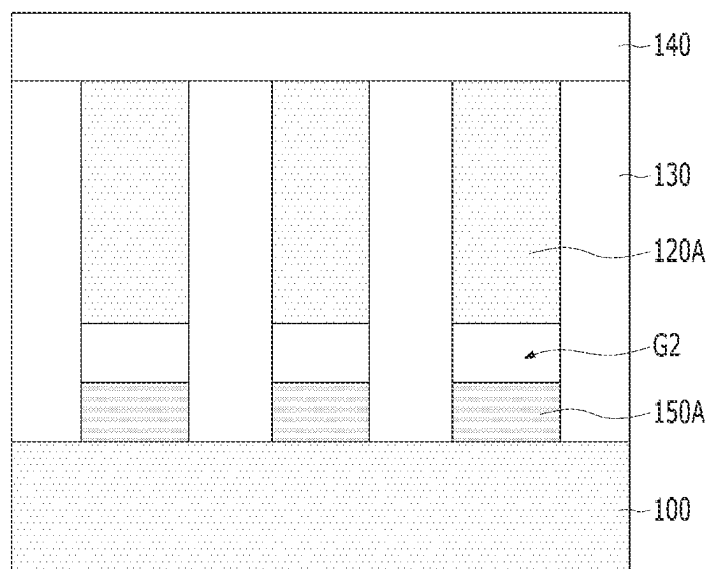


FIG. 6C

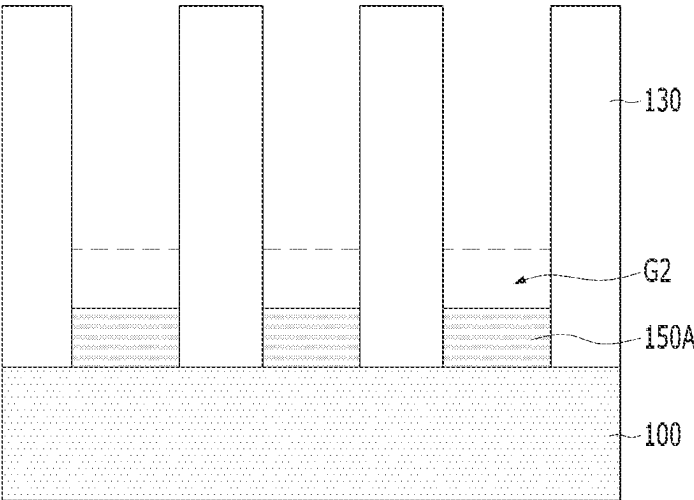


FIG. 6D

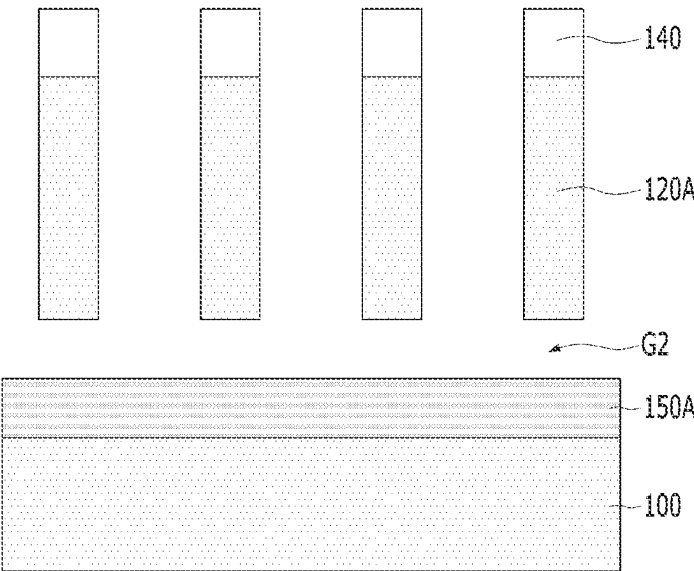


FIG. 7A

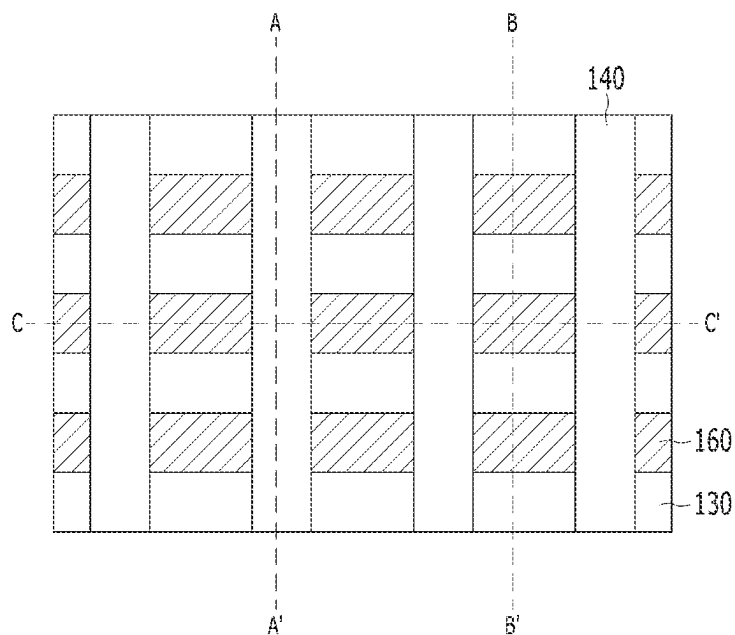


FIG. 7B

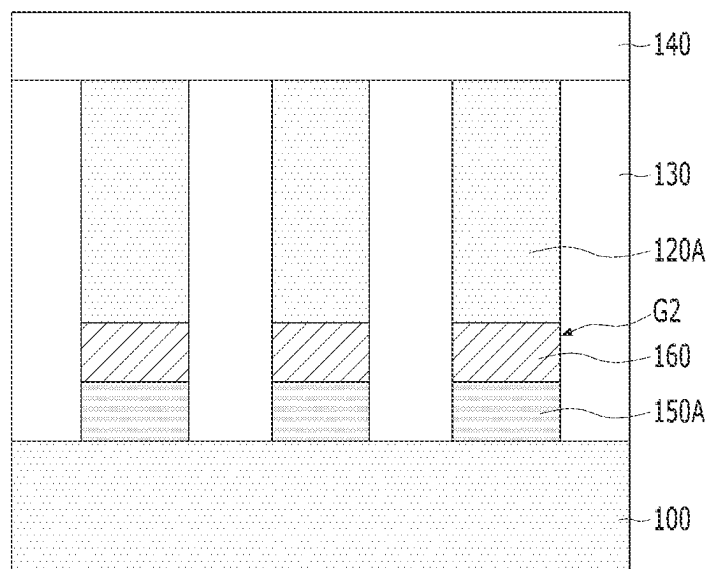


FIG. 7C

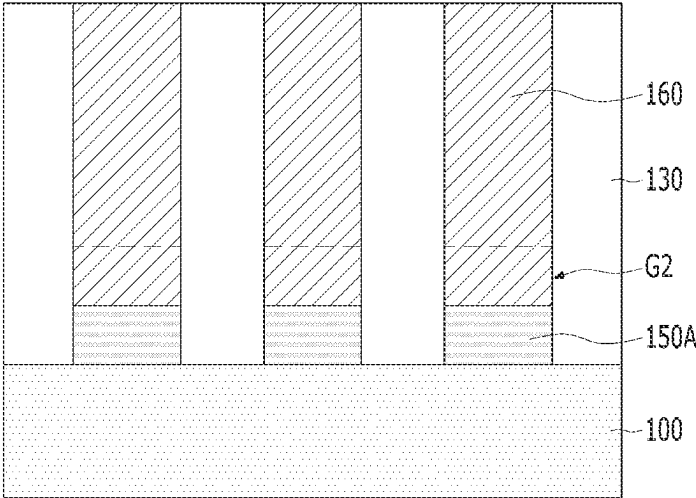


FIG. 7D

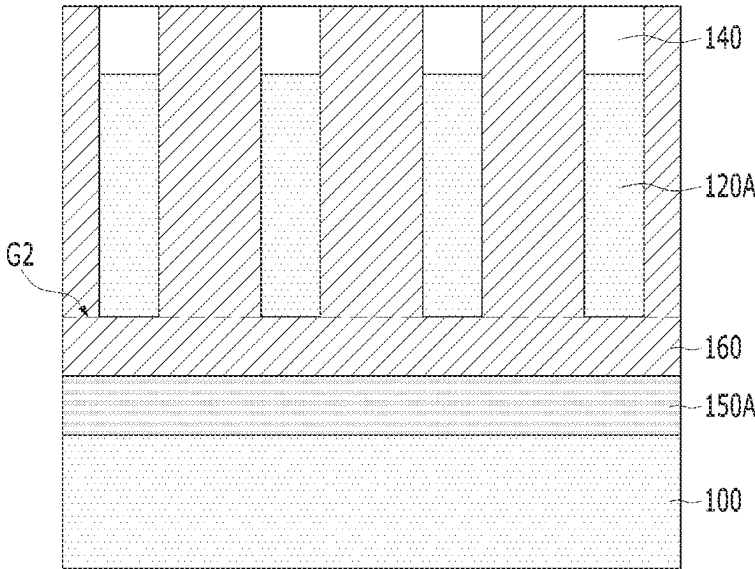


FIG. 8A

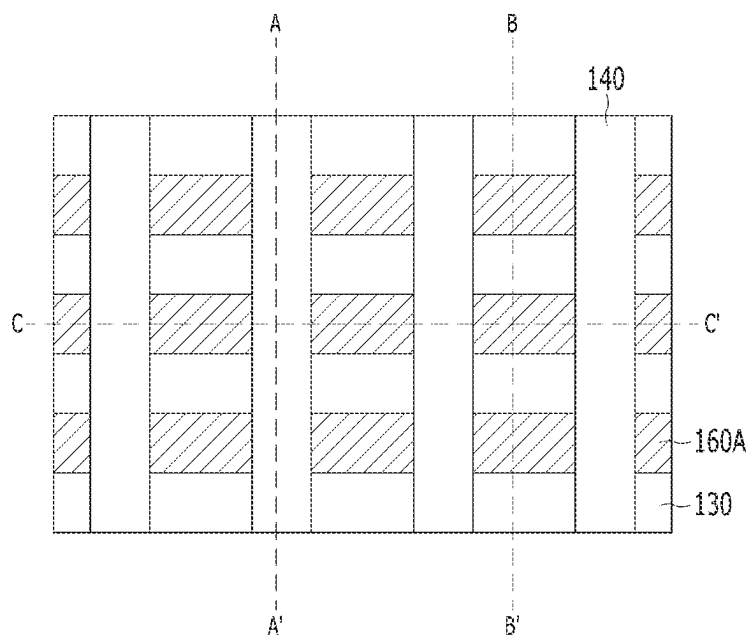


FIG. 8B

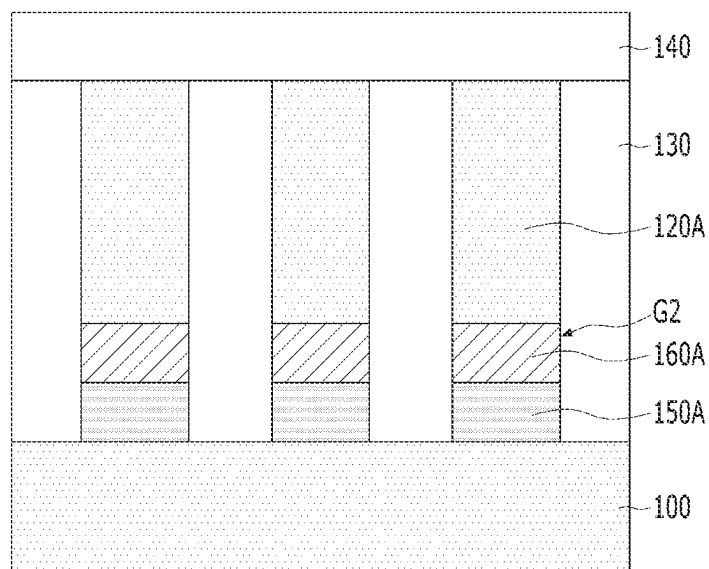


FIG. 8C

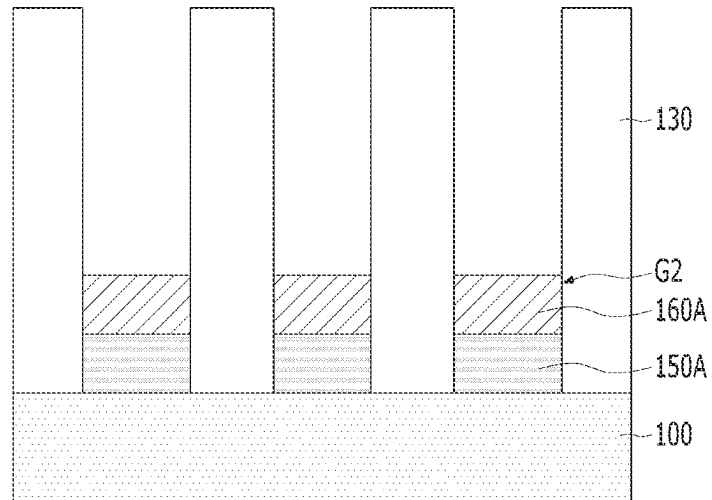


FIG. 8D

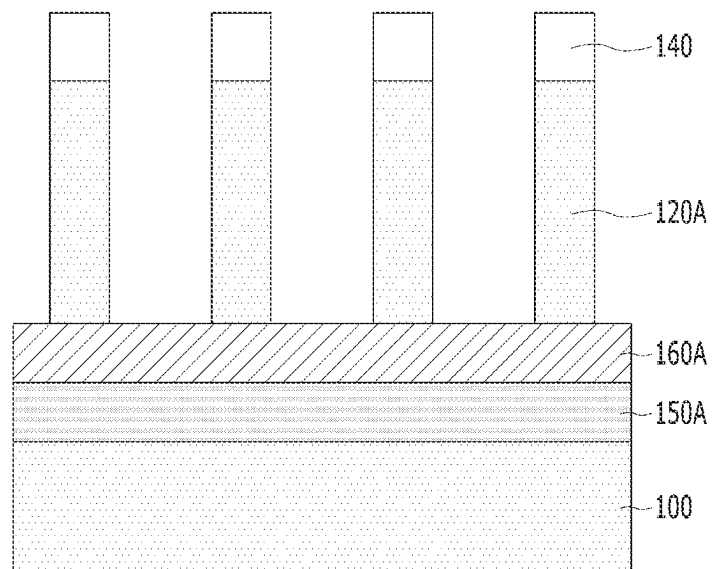


FIG. 9A

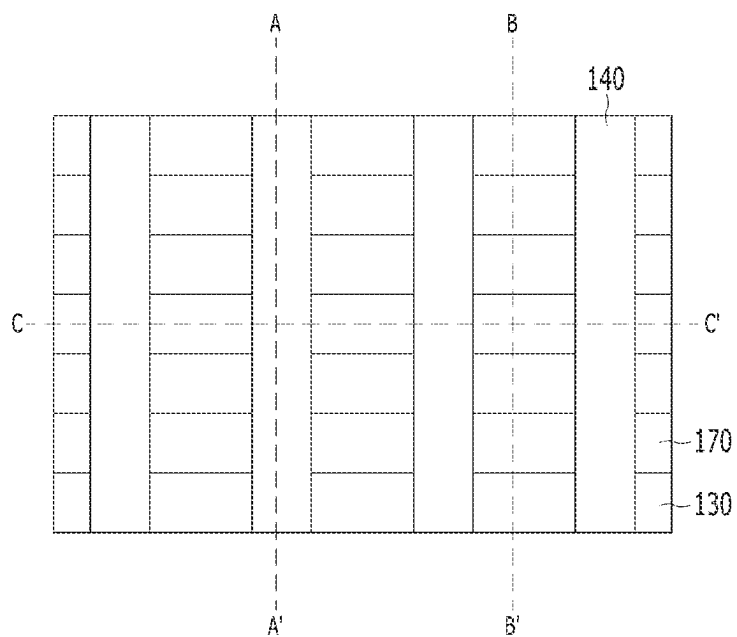


FIG. 9B

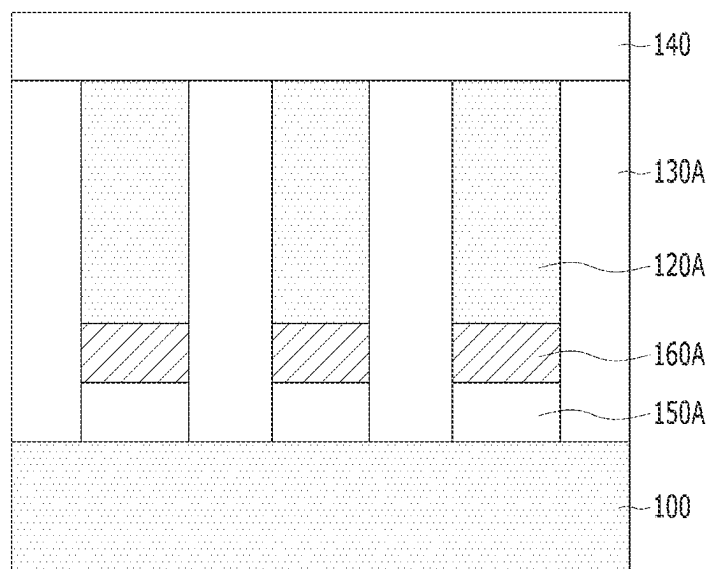


FIG. 9C

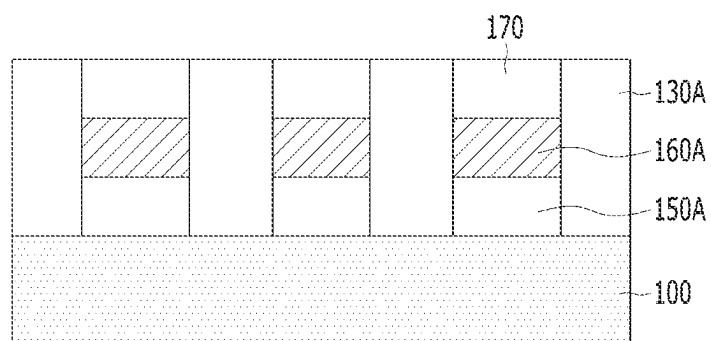


FIG. 9D

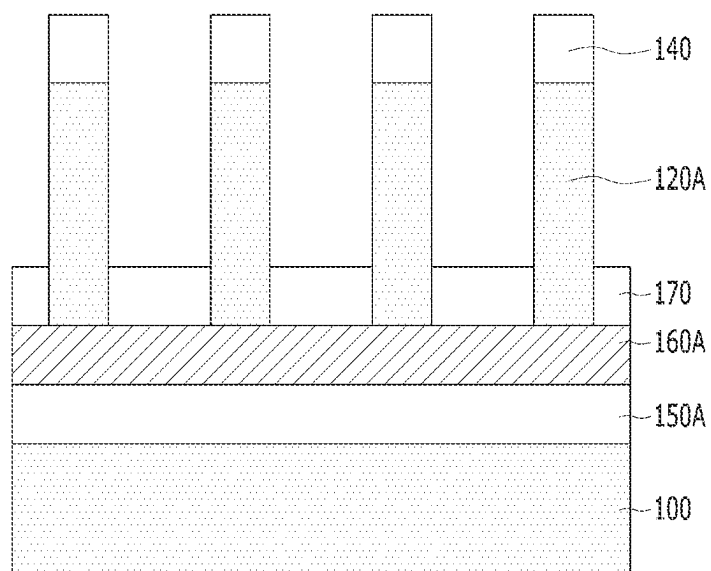


FIG. 10A

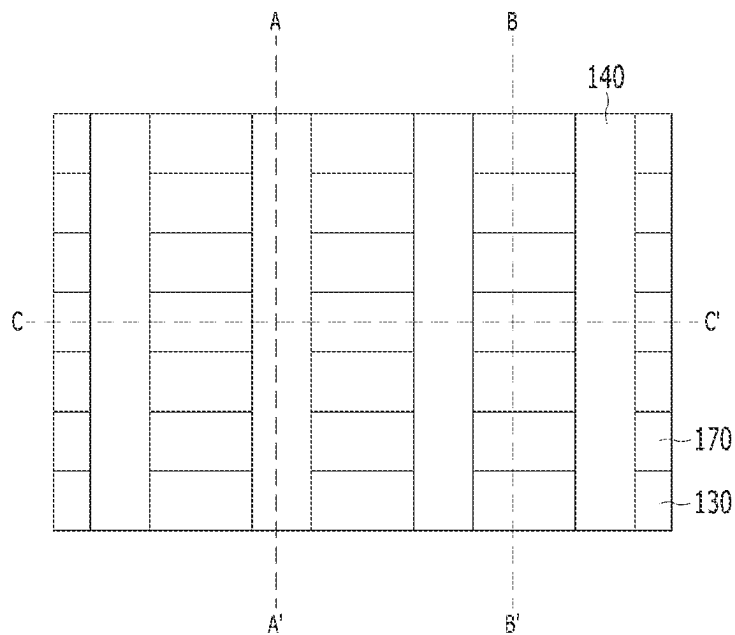


FIG. 10B

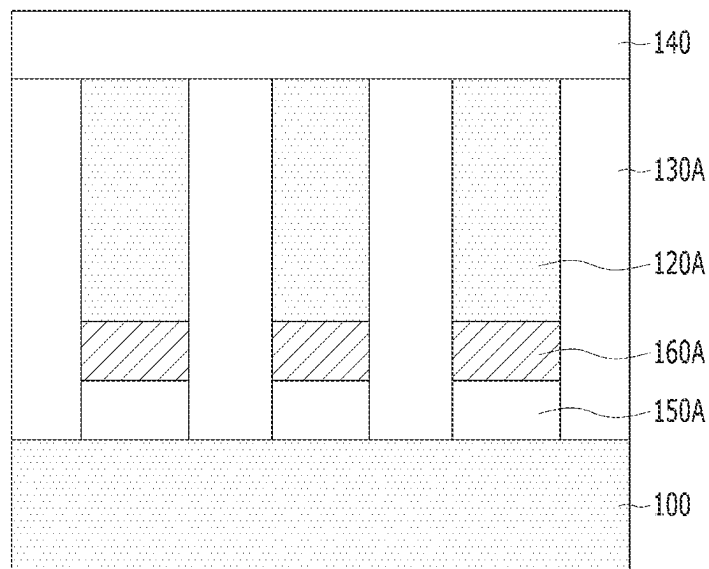


FIG. 10C

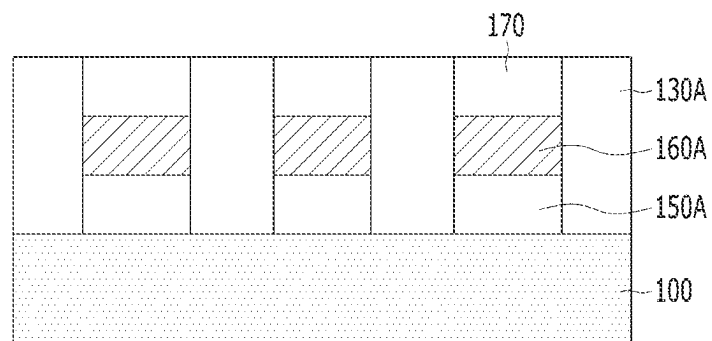


FIG. 10D

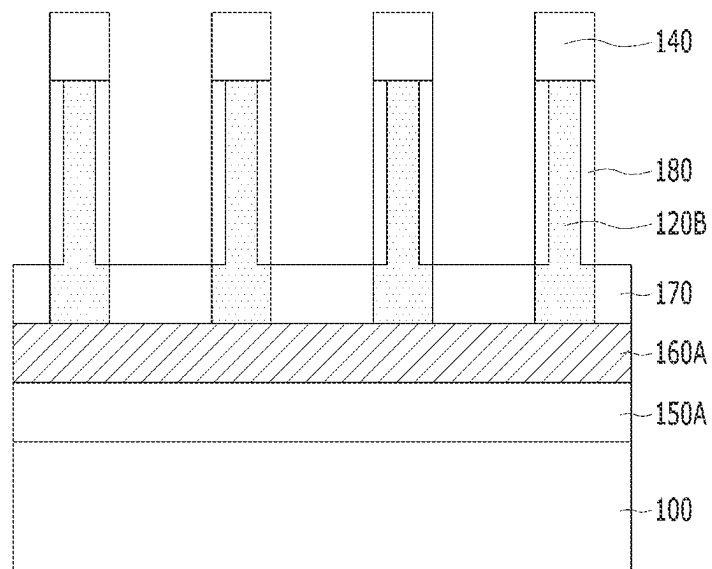


FIG. 11A

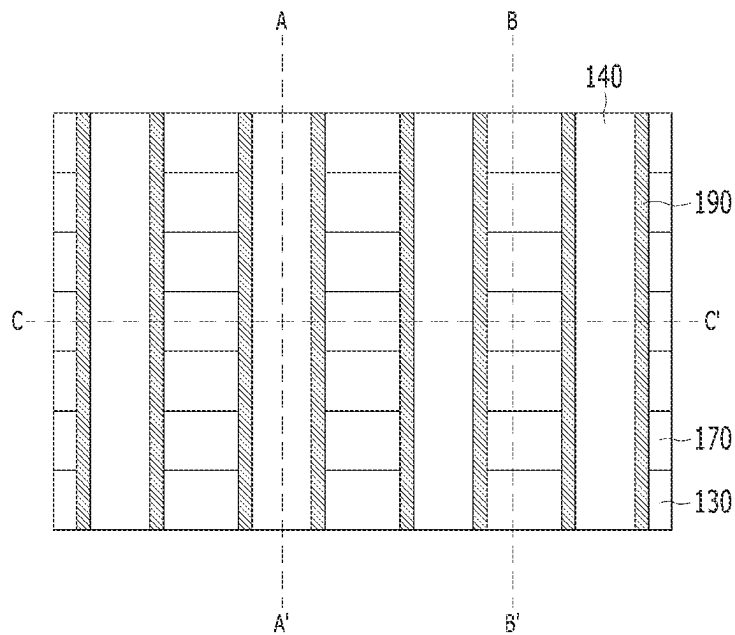


FIG. 11B

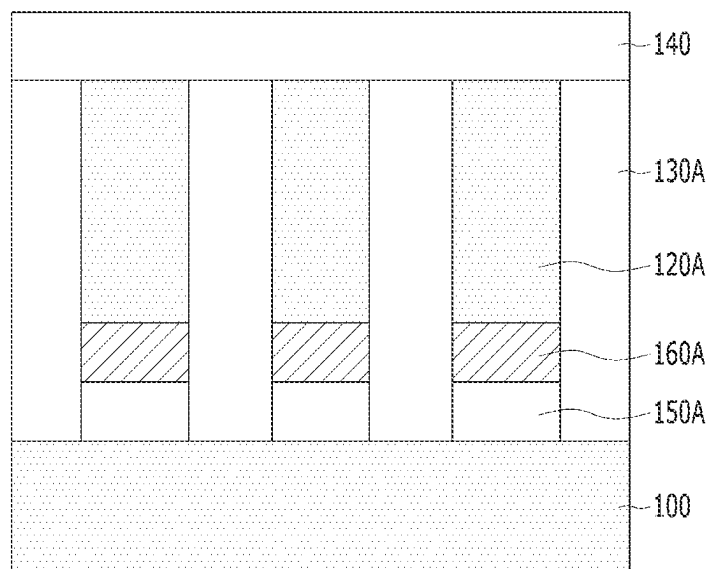


FIG. 11C

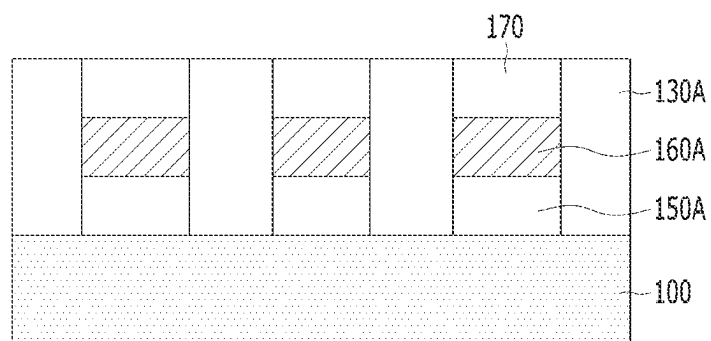


FIG. 11D

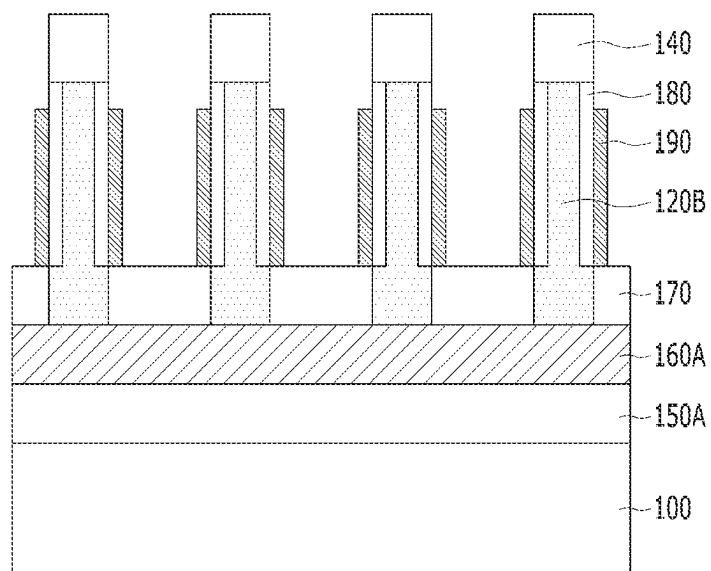


FIG. 12A

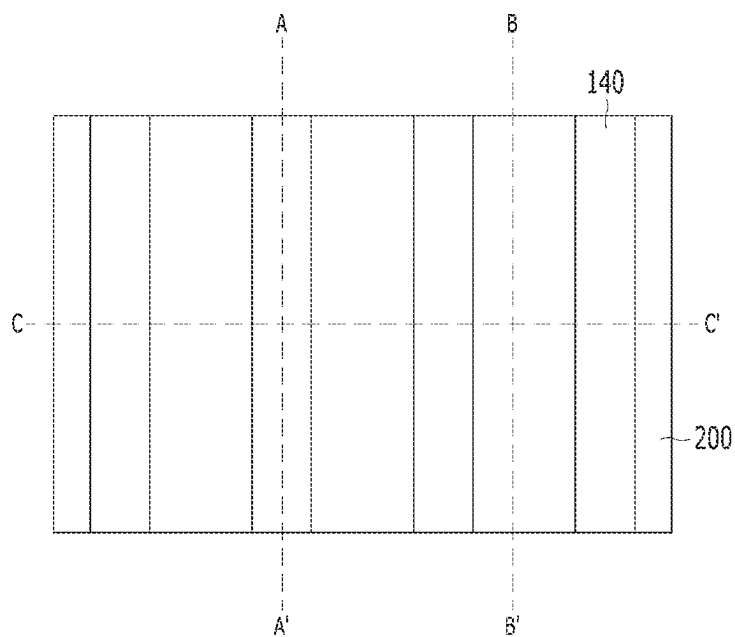


FIG. 12B

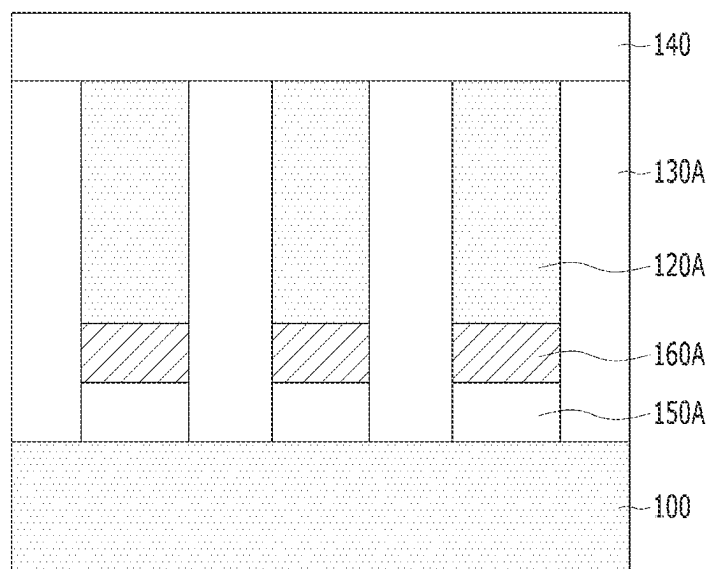


FIG. 12C

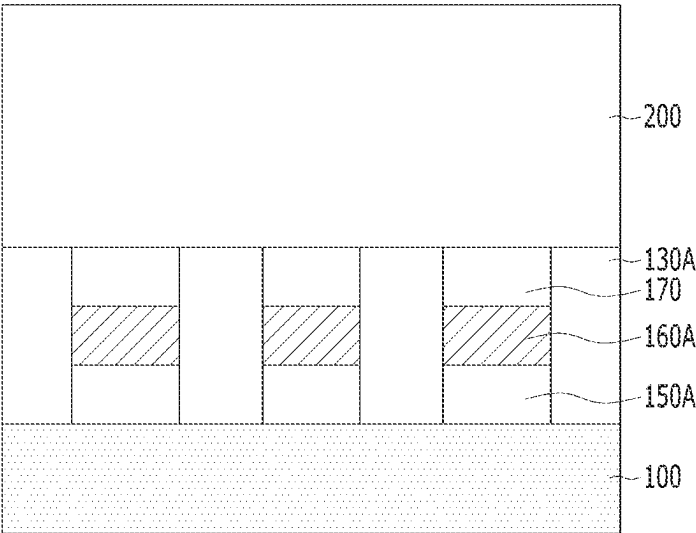


FIG. 12D

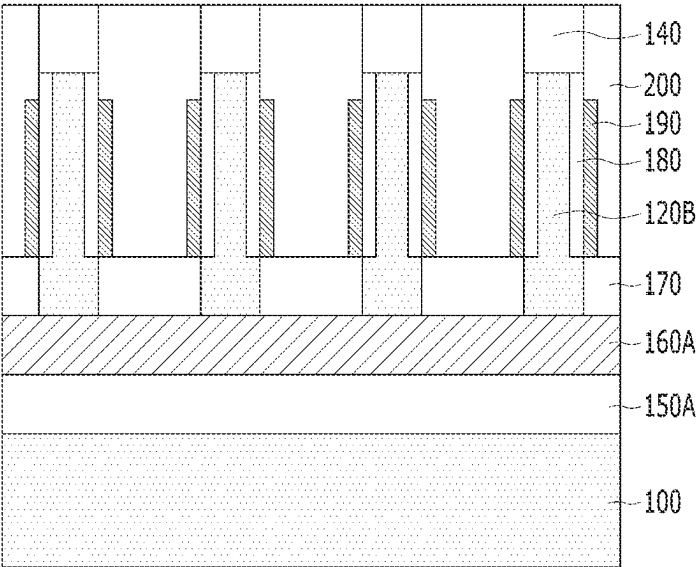


FIG. 13A

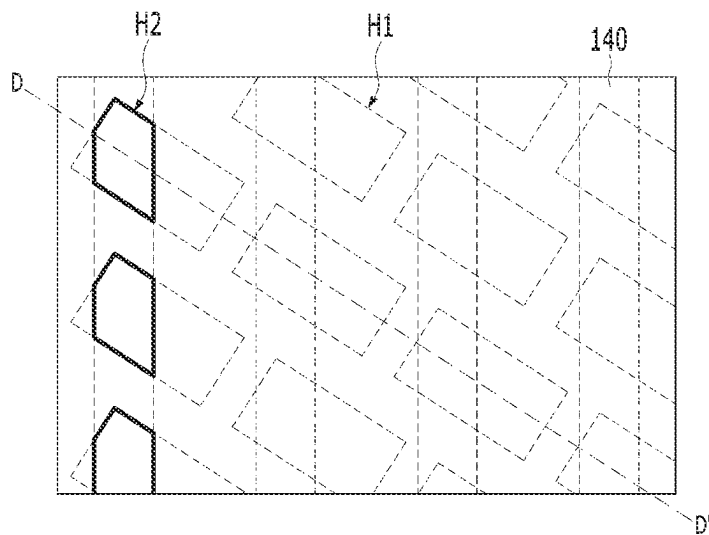


FIG. 13B

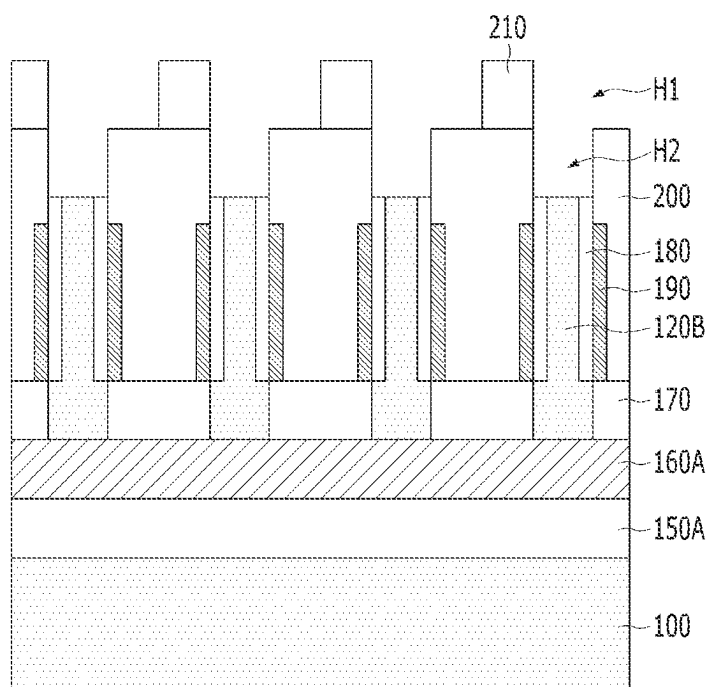


FIG. 14A

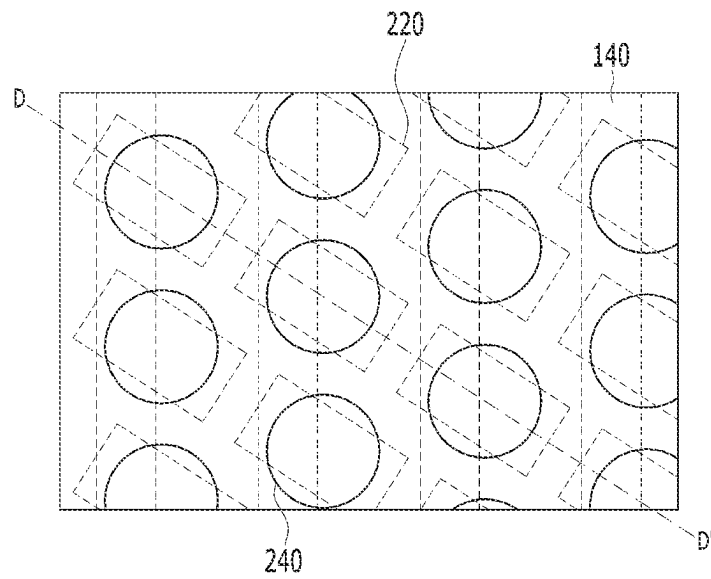


FIG. 14B

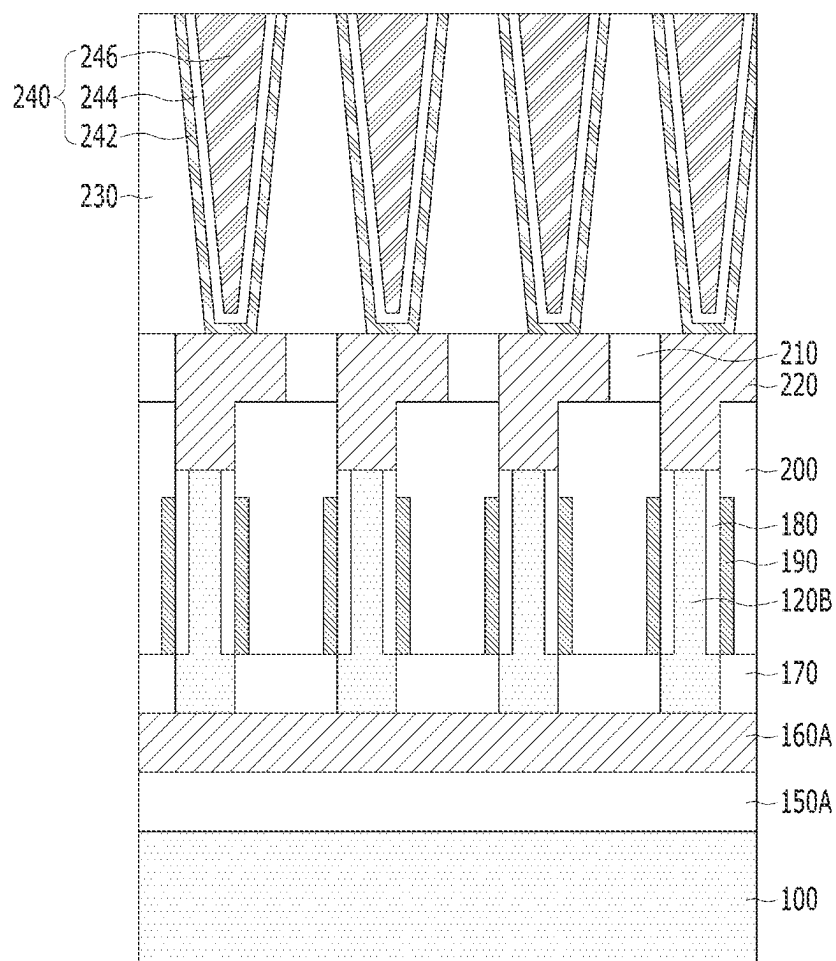


FIG. 15A

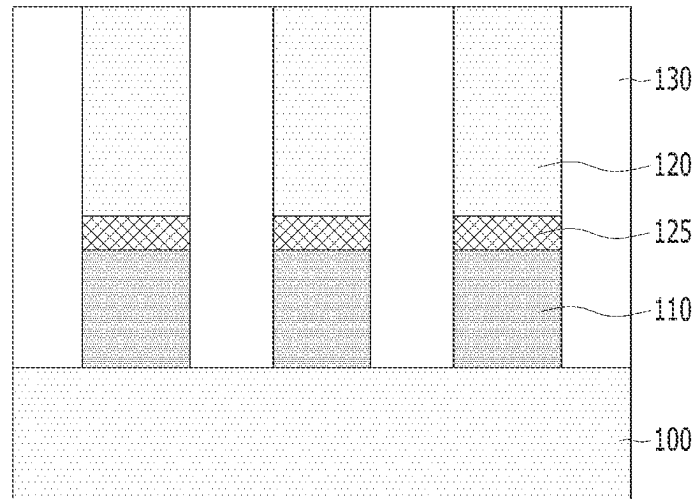


FIG. 15B

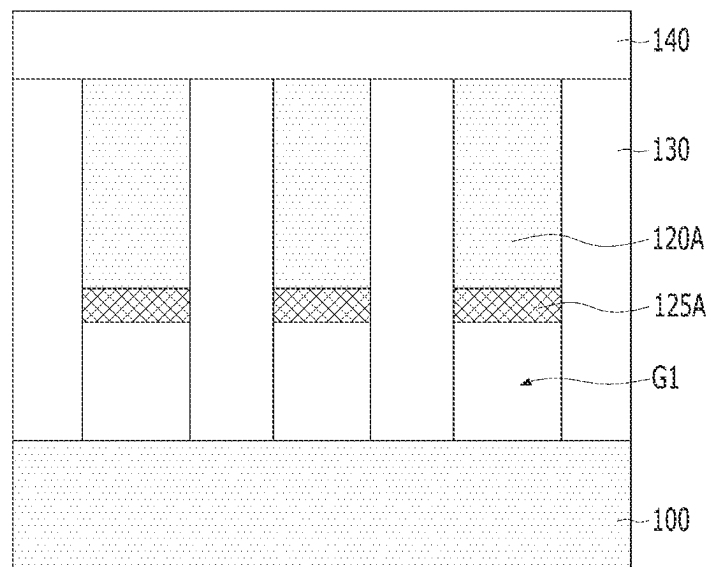
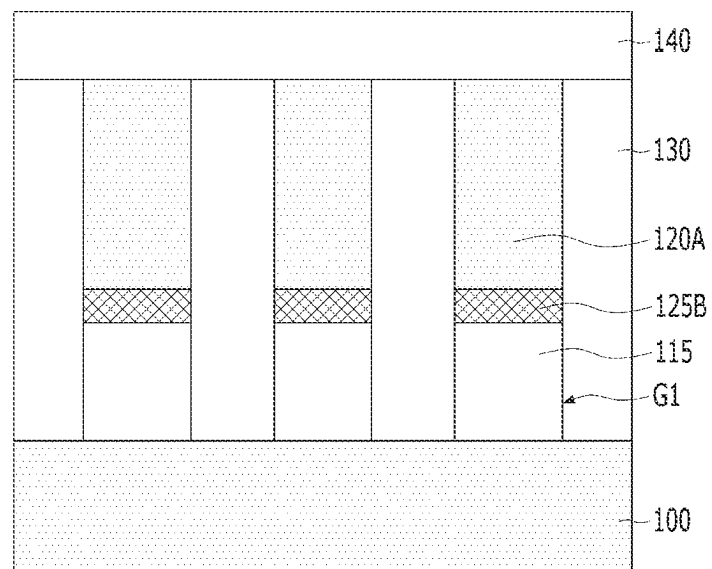


FIG. 15C



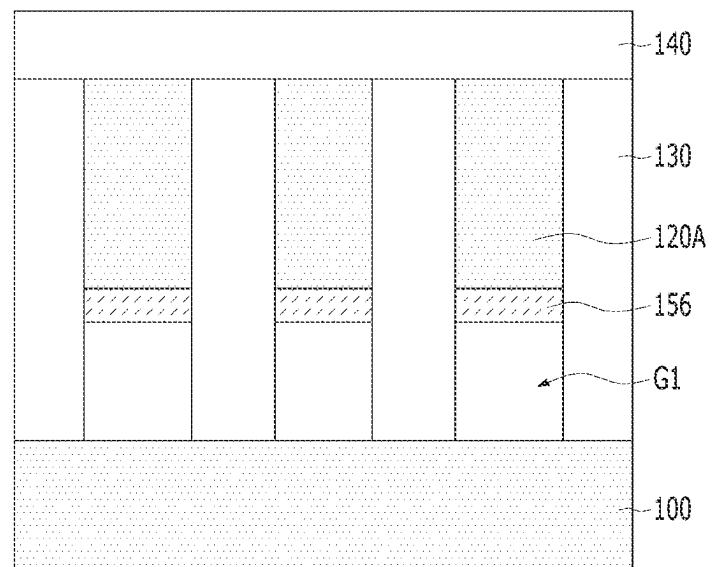


FIG. 17A

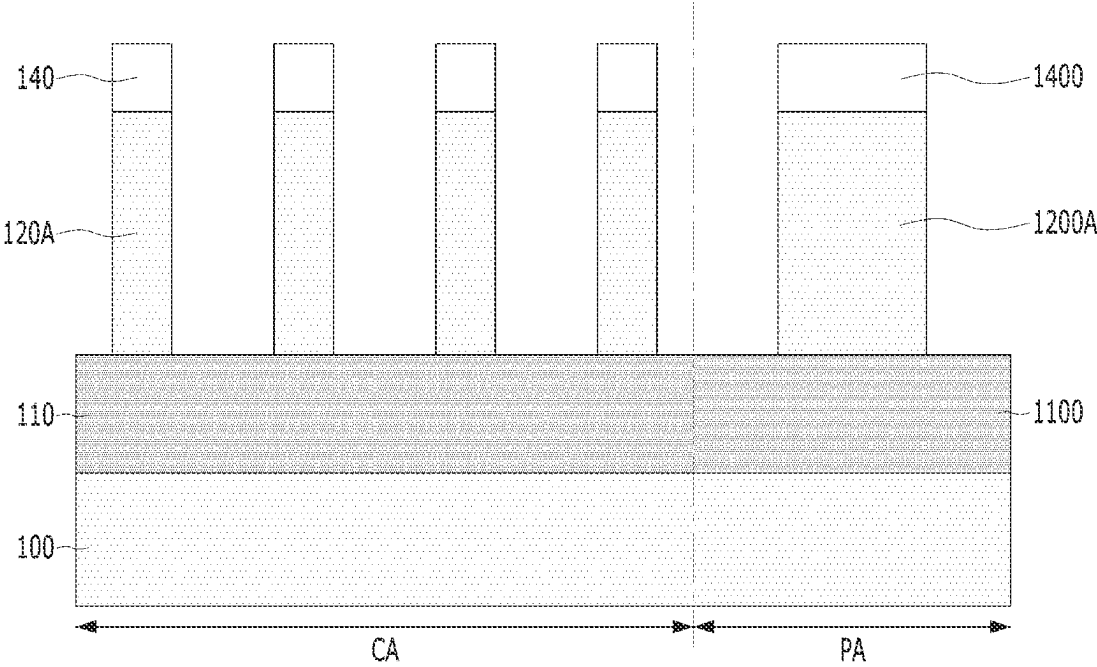


FIG. 17B

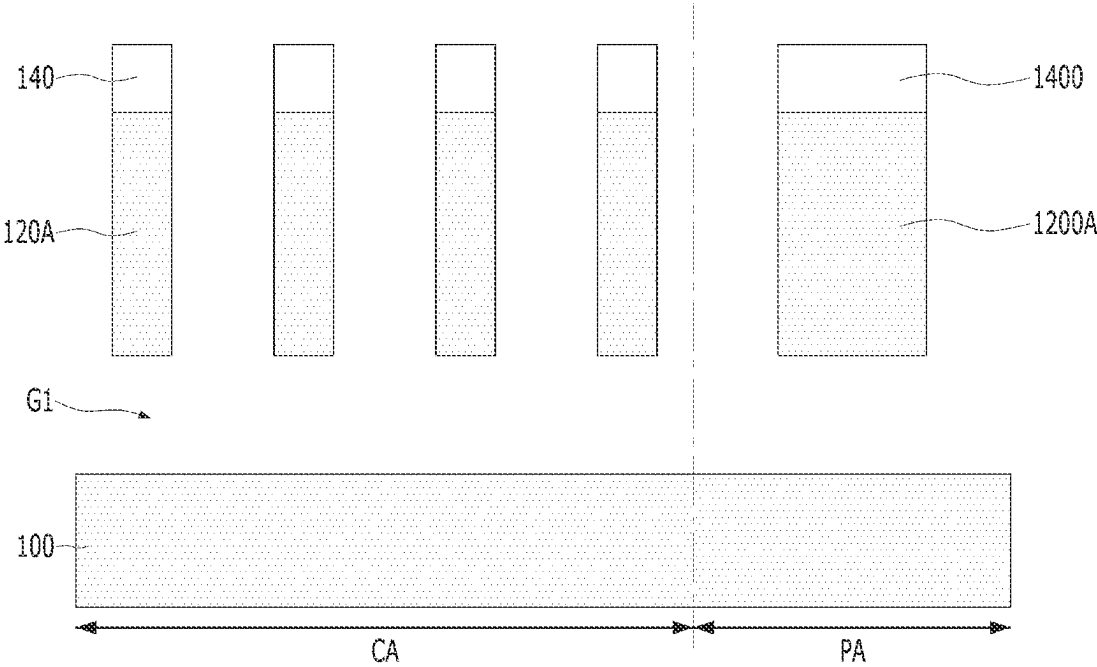


FIG. 17C

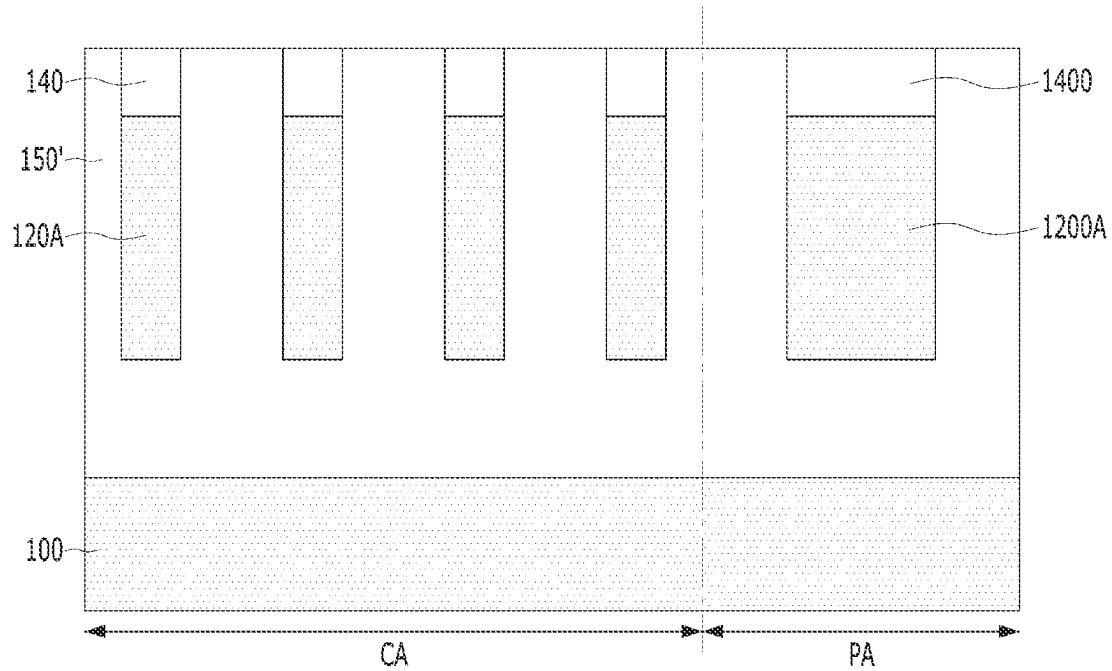


FIG. 17D

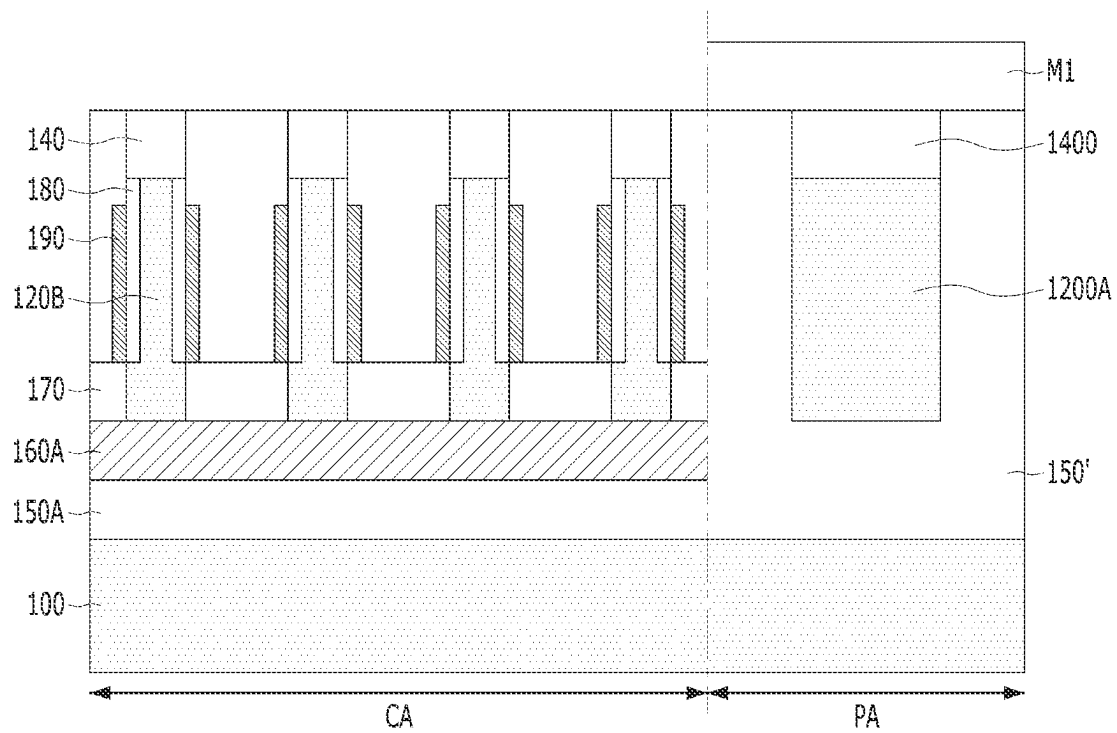


FIG. 17E

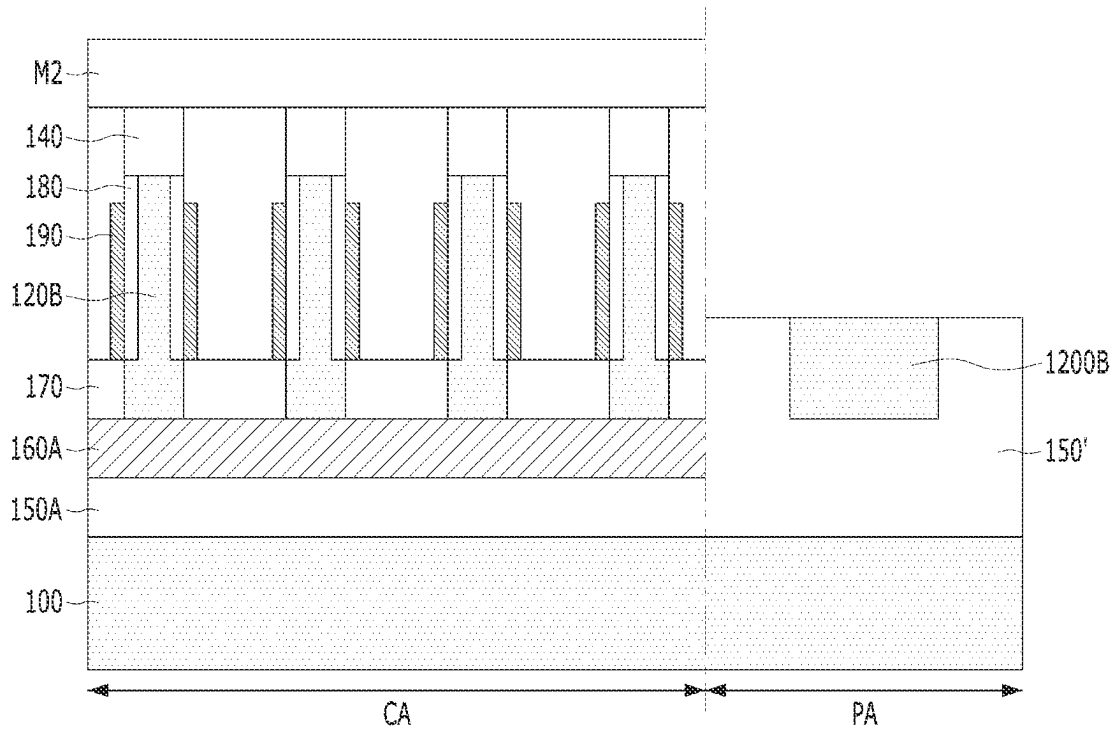
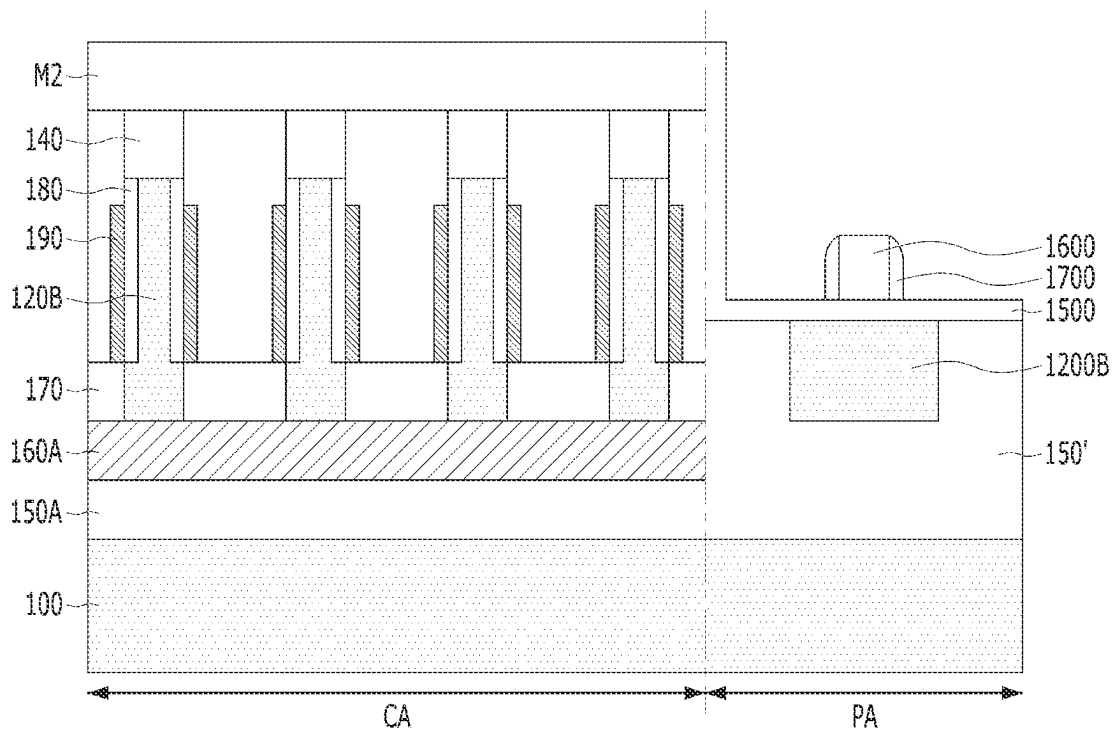


FIG. 17F



SEMICONDUCTOR DEVICE AND METHOD FOR FABRICATING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2022-0080774 filed on Jun. 30, 2022, which is incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

This patent document relates generally to semiconductor technology and, more particularly, to a semiconductor device including a transistor and a method for fabricating the same.

2. Related Art

Continuous development of the electronic industry requires electronic products which are gradually more miniaturized, have higher performance, are more highly integrated, and operate at higher speed. To satisfy this requirement, it is necessary to develop technologies capable of maintaining and/or improving the characteristics of the various unit elements of the semiconductor devices such as the transistors while reducing the size or footprint of the elements.

SUMMARY

In an embodiment, a semiconductor device may include: a substrate; a plurality of semiconductor pillars disposed over the substrate and arranged in a first direction and a second direction crossing the first direction; an insulating layer pattern disposed between the substrate and the semiconductor pillars and extending in the second direction; a first conductive line disposed between the insulating layer pattern and the semiconductor pillars and extending in the second direction; a second conductive line formed over sidewalls of the semiconductor pillars and extending in the first direction; and a storage node disposed over each of the semiconductor pillars.

In another embodiment, a method for fabricating semiconductor device, may include: forming a plurality of stacked structures extending in a second direction over a substrate, each of the stacked structures including a sacrificial layer and a semiconductor layer disposed over the sacrificial layer; forming a plurality of semiconductor pillars arranged in the second direction and a first direction crossing the second direction by selectively etching the semiconductor layer; forming a first groove by removing the sacrificial layer; forming an insulating layer pattern filling a bottom portion of the first groove; forming a first conductive line filling a remaining space of the first groove in which the insulating layer pattern is formed; forming a second conductive line extending in the first direction over sidewalls of the semiconductor pillars; and forming a storage node over each of the semiconductor pillars.

These and other features and advantages of the present invention will become apparent to the skilled person in this art from the following figures and detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A to 14B are schematic diagrams illustrating a semiconductor device according to an embodiment of the present disclosure, and a method for fabricating the same.

FIG. 15A is a schematic diagram illustrating an example of a method of forming a junction region at a bottom end of a semiconductor pillar pattern.

FIG. 15B is a schematic diagram illustrating another example of a method of forming a junction region at a bottom end of a semiconductor pillar pattern.

FIG. 15C is a schematic diagram illustrating still another example of a method of forming a junction region at a bottom end of a semiconductor pillar pattern.

FIGS. 16A and 16B are schematic diagrams illustrating an example of a method of forming a silicide layer under a semiconductor pillar pattern.

FIGS. 17A to 17F are schematic diagrams illustrating a semiconductor device according to another embodiment of the present disclosure, and a method for fabricating the same.

DETAILED DESCRIPTION

Hereinafter, various embodiments of the disclosure will be described in detail with reference to the accompanying drawings.

The drawings are not necessarily drawn to scale. In some instances, proportions of at least some structures in the drawings may have been exaggerated in order to clearly illustrate certain features of the described embodiments. In presenting a specific example in a drawing or description having two or more layers in a multi-layer structure, the relative positioning relationship of such layers or the sequence of arranging the layers as shown reflects a particular implementation for the described or illustrated example and a different relative positioning relationship or sequence of arranging the layers may be possible. In addition, a described or illustrated example of a multi-layer structure might not reflect all layers present in that particular multilayer structure (e.g., one or more additional layers may be present between two illustrated layers). As a specific example, when a first layer in a described or illustrated multi-layer structure is referred to as being “on” or “over” a second layer or “on” or “over” a substrate, the first layer may be directly formed on the second layer or the substrate but may also represent a structure where one or more other intermediate layers may exist between the first layer and the second layer or the substrate.

FIGS. 1A to 14B are schematic diagrams illustrating a semiconductor device according to an embodiment of the present disclosure, and a method for fabricating the same. FIGS. 1A, 2A, 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, and 14A are plan schematic diagrams from above. FIGS. 1B, 2B, 3B, 4B, 5B, 6B, 7B, 8B, 9B, 10B, 11B and 12B are cross-sectional schematic diagrams taken along lines A-A' of FIGS. 1A, 2A, 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, and 12A, respectively. FIGS. 1C, 2C, 3C, 4C, 5C, 6C, 7C, 8C, 9C, 10C, 11C, 12C, 13C, and 14C are cross-sectional schematic diagrams taken along lines B-B' of FIGS. 1A, 2A, 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, and 14A, respectively. FIGS. 1D, 2D, 3D, 4D, 5D, 6D, 7D, 8D, 9D, 10D, 11D, 12D, 13D, and 14D are cross-sectional schematic diagrams taken along lines C-C' of FIGS. 1A, 2A, 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, and 14A, respectively. FIGS. 13B and 14B are cross-

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sectional schematic diagrams taken along lines D-D' of FIGS. 13A and 14A, respectively.

First, a method for fabricating a semiconductor device of the present embodiment will be described.

Referring to FIGS. 1A to 1D, a stacked structure of a sacrificial layer 110 and a semiconductor layer 120 may be formed over a substrate 100.

The substrate 100 may include a semiconductor material such as, for example, silicon, and may have a plate shape in a plan view.

The stacked structure of the sacrificial layer 110 and the semiconductor layer 120 may have a line shape extending in a direction parallel to the line C-C' in a plan view. In addition, a plurality of stacked structures each including the sacrificial layer 110 and the semiconductor layer 120 may be arranged to be spaced apart from each other in a direction parallel to the line A-A' or the line B-B' in a plan view. In the following description, a direction parallel to the line A-A' or the line B-B' will be referred to as a first direction, and a direction parallel to the line C-C' will be referred to as a second direction. The first direction and the second direction may cross each other. For example, the first direction and the second direction may be perpendicular or substantially perpendicular to each other. The sacrificial layer 110 may include a material having an etch rate different from that of the substrate 100 and the semiconductor layer 120. For example, the sacrificial layer 110 may include SiGe. The semiconductor layer 120 may include a semiconductor material such as, for example, silicon, and may have a crystalline state. For example, the semiconductor layer 120 may include single crystal silicon.

The stacked structure of the sacrificial layer 110 and the semiconductor layer 120 may be formed by sequentially forming a sacrificial material for forming the sacrificial layer 110 and a semiconductor material for forming the semiconductor layer 120, over the substrate 100, and selectively etching the sacrificial material and the semiconductor material. Here, the sacrificial material for forming the sacrificial layer 110 may be formed by various deposition methods such as physical vapor deposition (PVD) and chemical vapor deposition (CVD), or epitaxial growth. In addition, the semiconductor material for forming the semiconductor layer 120 may be formed by various deposition methods such as, for example, PVD and CVD, or epitaxial growth.

Subsequently, a first insulating layer 130 filling a space between the stacked structures of the sacrificial layers 110 and the semiconductor layers 120 may be formed. The first insulating layer 130 may include various insulating materials such as, for example, silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof. Furthermore, the first insulating layer 130 may be formed of a material having an etch rate different from that of the sacrificial layer 120. The first insulating layer 130 may be formed by forming an insulating material covering the substrate 100 and the stacked structures of the sacrificial layers 110 and the semiconductor layers 120 until the top surface of the semiconductor layer 120 is exposed, and performing a planarization process, for example, chemical mechanical polishing (CMP) or etch-back.

Referring to FIGS. 2A to 2D, a hard mask layer 140 may be formed over the structure of FIGS. 1A to 1D.

The hard mask layer 140 may have a line shape extending in the first direction in a plan view. A plurality of hard mask layers 140 may be arranged to be spaced apart from each other in the second direction. The hard mask layer 140 may be used for patterning the semiconductor layer 120 into a pillar shape in a subsequent process while supporting a

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structure positioned thereunder when the sacrificial layer 110 is removed. The hard mask layer 140 may include a material having an etch rate different from that of the semiconductor layer 120 and the sacrificial layer 110. As an example, the hard mask layer 140 may include an insulating material such as, for example, silicon oxide, silicon nitride, silicon oxynitride, amorphous carbon, or a combination thereof.

Referring to FIGS. 3A to 3D, a semiconductor pillar 120A may be formed by etching the semiconductor layer 120 using the hard mask layer 140 as an etch barrier.

The semiconductor pillar 120A may overlap with each one of the intersection regions of the sacrificial layer 110 extending in the second direction and the hard mask layer 140 extending in the first direction between the sacrificial layer 110 and the hard mask layer 140, and may have a pillar shape, for example, a quadrangular pillar shape. Both sidewalls of the semiconductor pillar 120A in the first direction may be aligned with both sidewalls of the sacrificial layer 110, and both sidewalls of the semiconductor pillar 120A in the second direction may be aligned with both sidewalls of the hard mask layer 140.

In this etching process, the first insulating layer 130 and the sacrificial layer 110 exposed by the formation of the semiconductor pillar 120A may be maintained the same or substantially the same. For example, there is no loss due to etching or any loss due to the etching may be insubstantial.

Referring to FIGS. 4A to 4D, the sacrificial layer 110 may be removed to form a groove G1 in the space that is formed from the removal of the sacrificial layer 110.

The removal of the sacrificial layer 110 may be performed by an isotropic etching method, for example, a wet etching method. When the sacrificial layer 110 is removed, the substrate 100, the semiconductor pillar 120A, the first insulating layer 130, and the hard mask layer 140, which have an etch rate different from that of the sacrificial layer 110, may be maintained the same or substantially the same. Accordingly, the first groove G1 having the same or substantially the same shape as the removed sacrificial layer 110 may be formed. For example, the first groove G1 may have a line shape extending in the second direction, and may have the same or substantially the same width and thickness as the removed sacrificial layer 110.

Referring to FIGS. 5A to 5D, the second insulating layer 150 filling the first groove G1 may be formed.

The second insulating layer 150 may include various insulating materials such as, for example, silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof. Furthermore, the second insulating layer 150 may be formed of a material having an etch rate different from that of the substrate 100, the semiconductor pillar 120A, the first insulating layer 130, and the hard mask layer 140. The second insulating layer 150 may be formed by forming an insulating material to a thickness sufficient to fill the first groove G1 over the structure of FIGS. 4A to 4D, and removing the insulating material outside the first groove G1 by an etching process such as, for example, an etch-back process.

Referring to FIGS. 6A to 6D, a top portion (or upper portion) of the second insulating layer 150 may be removed to form a second insulating layer pattern 150A. In addition, a space formed by the removal of the top portion of the second insulating layer 150 may be referred to as a second groove G2. For example, the top portion of the second insulating layer 150 may be a portion of the second insulating layer 150 having a thickness smaller than that of the second insulating layer 150. The thickness of the top portion may be variously modified. For example, the top portion

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may be less than half or more than half of the total thickness of the second insulating layer **150**. A remaining portion of the second insulating layer **150** that is not removed and remains over the substrate **100** to a predetermined thickness, may form the second insulating layer pattern **150A**.

The removal of the top portion of the second insulating layer **150** may be performed, preferably, by an isotropic etching method. This etching process may be performed so that the second insulating layer pattern **150A** and the semiconductor pillar **120A** are completely separated by the second groove **G2** under the semiconductor pillar **120A**, and a portion of the second insulating layer **150** remains over the substrate **100** to a predetermined thickness. During the isotropic etching, undercut etching may start from just below the semiconductor pillar **120A** and etched portions may meet each other under the semiconductor pillar **120A**, and thus the second insulating layer pattern **150A** described above may be formed. When the second insulating layer pattern **150A** is formed, the substrate **100**, the semiconductor pillar **120A**, the first insulating layer **130**, and the hard mask layer **140**, which have an etch rate different from that of the second insulating layer pattern **150A**, may be maintained the same or substantially the same. Accordingly, the second insulating layer pattern **150A** and the second groove **G2**, which have the same or substantially the same shape as the sacrificial layer **110** in a plan view, may be formed. For example, the second insulating layer pattern **150A** and the second groove **G2** may have a line shape extending in the second direction, and may have the same width as the sacrificial layer **110**. However, since the second insulating layer pattern **150A** and the second groove **G2** occupy the space from which the sacrificial layer **110** is removed, the sum of the thicknesses of the second insulating layer pattern **150A** and the second groove **G2** is substantially equal to the thickness of the sacrificial layer **110**. Also, in these figures, the thickness of the second insulating layer pattern **150A** is drawn to be the same as the thickness of the second groove **G2**, but the present disclosure is not limited thereto. The thickness of the second insulating layer pattern **150A** may be smaller or greater than the thickness of the second groove **G2**.

Referring to FIGS. **7A** to **7D**, a first conductive layer **160** may be formed to fill the second groove **G2** and fill an empty space defined by the first insulating layer **130**, the semiconductor pillar **120A**, and the hard mask layer **140** over the second groove **G2**.

The first conductive layer **160** may be formed of various conductive materials, for example, a metal such as platinum (Pt), tungsten (W), aluminum (Al), copper (Cu), tantalum (Ta), titanium (Ti), ruthenium (Ru), or molybdenum (Mo), a compound of this metal, or an alloy of these metals. The first conductive layer **160** is formed by forming an insulating material to a thickness that covers the structure of FIG. **6A** to **6D** while sufficiently filling the second groove **G2**, and, performing a planarization process until the top surface of the hard mask layer **140** is exposed.

Referring to FIGS. **8A** to **8D**, a top portion of the first conductive layer **160** may be removed to form a first conductive line **160A** filled in the second groove **G2**.

The top portion of the first conductive layer **160** may be removed by an etching process such as, for example, an etch-back process. The first conductive line **160A** may have the same or substantially the same shape, width, and thickness as the second groove **G2**. For example, the first conductive line **160A** may be disposed over the second insulating layer pattern **150A**, and may overlap with the second insulating layer pattern **150A** in a plan view to have the same

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or substantially the same shape and width as the second insulating layer pattern **150A**. In the first direction, both sidewalls of the first conductive line **160A** may be aligned with both sidewalls of the second insulating layer pattern **150A**. The first conductive line **160A** may be electrically and physically separated from the substrate **100** by the second insulating layer pattern **150A**. Furthermore, a plurality of first conductive lines **160A** may be arranged to be spaced apart from each other in the first direction, and may be electrically and physically separated from each other by the first insulating layer **130**. The first conductive line **160A** may function as a bit line electrically connected to one terminal of a transistor, for example, a drain terminal.

Referring to FIGS. **9A** to **9D**, an insulating material having a thickness sufficient to fill an empty space defined by the first insulating layer **130**, the semiconductor pillar **120A**, and the hard mask layer **140** may be formed over the structure of FIGS. **8A** and **8B**, and an etching process such as, for example, an etch-back process may be performed so that the insulating material remains over the first conductive line **160A** to a predetermined thickness. Accordingly, a third insulating layer **170** may be formed to cover a portion of the first conductive line **160A**, which is not covered by the hard mask layer **140**.

Here, the third insulating layer **170** may include various insulating materials such as, for example, silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof. The third insulating layer **170** may be used to electrically and physically separate the first conductive line **160A** from a second conductive line (refer to **190** of FIGS. **11A** to **11D**) to be formed in a subsequent process. In order to provide a space in which the second conductive line is to be formed, the third insulating layer **170** may have a top surface that is lowered by a predetermined degree from the top surface of the semiconductor pillar **120A**. Accordingly, a portion of the semiconductor pillar **120A** may protrude upward from the third insulating layer **170**. The portion of the semiconductor pillar **120A** that protrudes above the third insulating layer **170** will be hereinafter referred to as a protruding portion of the semiconductor pillar **120A**.

In an embodiment, where the third insulating layer **170** includes the same insulating material as the first insulating layer **130**, a portion of the first insulating layer **130**, which is not covered by the hard mask layer **140** and exposed, may be etched during the forming process of the third insulating layer **170**, and thus, a first insulating layer pattern **130A** may be formed. The first insulating layer pattern **130A** may include a first portion positioned under the hard mask layer **140** and a second portion not covered by the hard mask layer **140**. The first portion may have a top surface having the same or substantially the same height as the top surface of the semiconductor pillar **120A**, and the second portion may have a top surface having the same or substantially the same height as a top surface of the third insulating layer **170**.

Referring to FIGS. **10A** to **10D**, a gate insulating layer **180** may be formed along a surface of the protruding portion of the semiconductor pillar **120A**.

In the present embodiment, the gate insulating layer **180** may be formed by oxidizing a portion from a side surface of the protruding portion of the semiconductor pillar **120A**. In this case, a width of the protruding portion of the semiconductor pillar **120A** may be reduced so that a semiconductor pillar pattern **120B** is formed, and the gate insulating layer **180** may include an oxide of a material constituting the semiconductor pillar **120A**, for example, silicon oxide. However, the present disclosure is not limited thereto, and the gate insulating layer **180** may be formed over the side

surface of the protruding portion of the semiconductor pillar **120A** by various deposition methods, and may include various insulating materials such as, for example, silicon oxide, silicon nitride, silicon oxynitride, or the like, or a high-k material such as aluminum oxide. When the gate insulating layer **180** is formed over the side surface of the semiconductor pillar **120A** by a deposition method, the semiconductor pillar **120A** may be maintained the same or substantially the same. For example, the width of the semiconductor pillar **120A** after the gate insulating layer **180** is formed may be substantially the same as the width of the semiconductor pillar **120A** before the gate insulating layer **180** is formed.

Referring to FIGS. **11A** to **11D**, a second conductive lines **190** extending in the first direction may be formed. Two second conductive lines **190** may be formed over both sidewalls of the gate insulating layer **180** in the second direction, respectively.

Each of the second conductive lines **190** may be formed by depositing a conductive material conformally along a lower profile over the structure of FIGS. **10A** to **10D**, and performing blanket etching. The blanket etching may be performed to expose the surface of the third insulating layer **170** so that the adjacent second conductive lines **190** in the second direction are separated from each other. For reference, in the structure of FIGS. **10A** to **10D**, the first insulating layer patterns **130A** and the semiconductor pillar patterns **120B** in which the gate insulating layer **180** is formed may be alternately arranged along the first direction, under the hard mask layer **140** extending in the first direction. Accordingly, the second conductive lines **190** may extend in the first direction to be formed over both sidewalls of the first insulating layer patterns **130A** and the gate insulating layers **180** alternately arranged in the first direction.

The second conductive line **190** may include various conductive materials, for example, a metal such as platinum (Pt), tungsten (W), aluminum (Al), copper (Cu), tantalum (Ta), titanium (Ti), ruthenium (Ru), or molybdenum (Mo), a compound of this metal, or an alloy of these metals. The second conductive line **190** may function as a word line electrically connected to a gate terminal of a transistor.

Referring to FIGS. **12A** to **12D**, an insulating material covering the structure of FIGS. **11A** to **11D** may be formed, and a planarization process may be performed until the hard mask layer **140** is exposed. Accordingly, a fourth insulating layer **200** may be formed. The fourth insulating layer **200** may include various insulating materials such as, for example, silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof. Furthermore, the fourth insulating layer **200** may include an insulating material having an etch rate different from that of the hard mask layer **140**.

Referring to FIGS. **13A** and **13B**, a fifth insulating layer **210** may be formed over the hard mask layer **140** and the fourth insulating layer **200**. The fifth insulating layer **210** may be selectively etched to form a first hole **H1** exposing a portion of the first mask layer **140**.

The fifth insulating layer **210** may include an insulating material having an etch rate different from that of the hard mask layer **140**. As an example, the fifth insulating layer **210** may be formed of the same material as the fourth insulating layer **200**. In a plan view, the first hole **H1** may be disposed to overlap with each of the plurality of semiconductor pillar patterns **120B** under the hard mask layer **140**, and may have a larger planar area than the semiconductor pillar pattern **120B**. Accordingly, the first hole **H1** may partially overlap with the fourth insulating layer **200**. This allows securing a

space for forming a storage node as described later. However, in this case, since a sufficient distance must exist between the adjacent first holes **H1**, the first hole **H1** may have a bar shape having a long side and a short side, or a shape similar thereto. For example, the first hole **H1** may have a long side which is parallel to the line **D-D'**. A direction parallel to the line **D-D'**, for example, a diagonal direction with respect to the first direction and the second direction, will be referred to as a third direction.

Next, by removing a portion of the hard mask layer **140** exposed by the first hole **H1**, a second hole **H2** exposing the top surface of the semiconductor pillar pattern **120B** may be formed in the hard mask layer **140**. The second hole **H2** partially overlaps with the first hole **H1**, and the first and second holes **H1** and **H2** are integrally connected to each other.

Referring to FIGS. **14A** and **14B**, a conductive pattern **220** may be formed by filling the first and second holes **H1** and **H2**.

The conductive pattern **220** may be formed by first applying a conductive material having a thickness sufficiently to fill the first and second holes **H1** and **H2** over the structure of FIGS. **13A** and **13B**, and then performing a planarization process until the top surface of the fifth insulating layer **210** is exposed. The conductive pattern **220** may include various conductive materials, for example, a metal such as platinum (Pt), tungsten (W), aluminum (Al), copper (Cu), tantalum (Ta), titanium (Ti), ruthenium (Ru), or molybdenum (Mo), a compounds of this metal, or an alloy of these metals. The conductive pattern **220** may correspond to a storage node contact electrically connected to the other terminal of a transistor, for example, a source terminal.

Subsequently, a storage node **240** electrically connected to the conductive pattern **220** may be formed over the conductive pattern **220**. Since the area of the top surface of the conductive pattern **220** is larger than the area of the top surface of the semiconductor pillar pattern **120B** in a plan schematic diagram, the contact characteristics between the conductive pattern **220** and the storage node **240** may be improved.

The storage node **240** may include a capacitor with a dielectric layer **244** interposed between two electrodes **242** and **246**. In the present embodiment, the storage node **240** may be formed by forming a sixth insulating layer **230** over the conductive pattern **220** and the fifth insulating layer **210**, selectively etching the sixth insulating layer **230** to form a hole exposing the conductive pattern **220**, depositing a conductive material along an inner wall of the hole to form a first electrode **242**, forming the dielectric layer **244** over the first electrode **242** along its surface, and filling a remaining space of the hole where the first electrode **242** and the dielectric layer **244** are formed with a conductive material to form a second electrode **246**. In this case, a cylindrical capacitor may be formed.

However, the present disclosure is not limited thereto, and the storage node **240** may include capacitors having various shapes. Alternatively, the storage node **240** may include various devices capable of storing data, instead of the capacitor. As an example, the storage node **240** may include a variable resistance element that stores different data by switching in different resistance states.

The semiconductor device of the present embodiment may be fabricated by the fabricating method described above.

The semiconductor device of the present embodiment may include the substrate **100**, a stacked structure of the second insulating layer pattern **150A** and the first conductive

line 160A disposed over the substrate 100 and extending in the second direction, the semiconductor pillar patterns 120 disposed over the first conductive line 160A and arranged along the first and second directions, the second conductive line 190 formed over a portion of the semiconductor pillar pattern 120B with the gate insulating layer 180 interposed therebetween, in particular, the sidewall of the protruding portion of the semiconductor pillar 120B and extending in the first direction, and the storage node 240 formed over the semiconductor pillar pattern 120B and electrically connected to the semiconductor pillar pattern 120B through the conductive pattern 220.

The semiconductor pillar pattern 120B may function as a channel of a transistor, and may be controlled by the second conductive line 190 serving as a word line. A bottom end of the semiconductor pillar pattern 120B may function as a drain terminal of the transistor, and may be connected to the first conductive line 160A serving as a bit line. A top end of the semiconductor pillar pattern 120B may function as a source terminal of the transistor, and may be connected to the storage node 240. Accordingly, the semiconductor device including the transistor having the channel extending in a vertical direction, and the storage node 240, the word line, and the bit line which are connected to the transistor may be implemented.

A detailed description of each of the components of the semiconductor device of the present embodiment has already been described in the process of describing the above-described fabricating method, and thus, the detailed description will be omitted.

According to the semiconductor device and the fabricating method thereof in the present embodiment, the following advantages may be obtained.

First, since the semiconductor pillar pattern 120B used as the channel of the transistor extends in the vertical direction and the first conductive line 160A used as the bit line is filled thereunder, the area of the semiconductor device may be reduced.

In addition, since the semiconductor pillar pattern 120B is formed independently from the substrate 100 by the first conductive line 160A, it may be facilitated to form the semiconductor pillar pattern 120B having a desired characteristic. The semiconductor pillar pattern 120B and the substrate 100 may have different semiconductor materials and/or crystal structures. As an example, the semiconductor pillar pattern 120B may include a single crystal semiconductor, for example, single crystal silicon, which has an excellent characteristic, for example, improved current transfer capability.

In addition, by interposing the second insulating layer pattern 150A between the first conductive line 160A and the substrate 100, mutual interference such as leakage occurring between the first conductive lines 160A through the substrate 100 may be prevented or reduced.

As a result, the degree of integration and operating characteristics of the semiconductor device may be improved.

Meanwhile, the bottom and top ends of the aforementioned semiconductor pillar pattern 120B may function as the drain terminal and the source terminal, respectively. To this end, it may be necessary to form a junction region by doping with impurities at each of the bottom and top ends of the semiconductor pillar pattern 120B. Since the top end of the semiconductor pillar pattern 120B is exposed through the first and second holes H1 and H2 formed by the process of FIGS. 13A and 13B, impurities may be doped by ion implantation toward the top end of the exposed semicon-

ductor pillar pattern 120B before filling the first and second holes H1 and H2 with the conductive pattern 220. Thus, forming of the junction region may be facilitated at the top end of the semiconductor pillar pattern 120B. A method of forming the junction region at the bottom end of the semiconductor pillar pattern 120B will be described by way of example with reference to FIGS. 15A to 15C below.

FIG. 15A is a schematic diagram illustrating an example of a method of forming a junction region at a bottom end of a semiconductor pillar pattern, FIG. 15B is a schematic diagram illustrating another example of a method of forming a junction region at a bottom end of a semiconductor pillar pattern, and FIG. 15C is a schematic diagram illustrating still another example of a method of forming a junction region at a bottom end of a semiconductor pillar pattern.

Referring to FIG. 15A, the sacrificial layer 110 may be formed of a material doped with impurities, for example, SiGe doped with impurities.

In this case, the impurities of the sacrificial layer 110 may diffuse to the bottom end of the semiconductor layer 120 to be transformed into a semiconductor pillar pattern to form a junction region 125.

Alternatively, referring to FIG. 15B, an isotropic doping process, for example, an isotropic plasma doping process, may be performed in a state in which the first groove G1 is formed.

In this case, impurities may be doped into the bottom end of the semiconductor pillar 120A exposed by the first groove G1 to form a junction region 125A.

Alternatively, referring to FIG. 15C, a layer 115 doped with impurities, for example, a glass layer doped with impurities, may be formed in a state in which the first groove G1 is formed, and the impurities of the layer 115 may diffuse to the bottom end of the semiconductor pillar 120A by a heat treatment to form a junction region 125B.

After forming the junction region 125B, the layer 115 may be removed, and subsequent processes, such as the processes of FIGS. 5A and 5B and subsequent processes thereof, may be performed.

In the embodiments of FIGS. 15B and 15C, the junction regions 125A and 125B may be formed after the first groove G1 is formed, but the present disclosure is not limited thereto. In another embodiment, after the second groove G2 is formed (see FIGS. 6A and 6B), a junction region may be formed in the same manner.

In addition, a silicide layer may be formed to reduce contact resistance between the above-described semiconductor pillar pattern 120B and the first conductive line 160A and/or between the semiconductor pillar pattern 120B and the conductive pattern 220. Since the top end of the semiconductor pillar pattern 120B is exposed through the first and second holes H1 and H2 formed by the process of FIGS. 13A and 13B, a silicide layer may be formed over the semiconductor pillar pattern 120B by depositing a metal material over the top surface of the semiconductor pillar pattern 120B and performing a silicide process. A method of forming a silicide layer under the semiconductor pillar pattern 120B will be described by way of example with reference to FIGS. 16A and 16B below.

FIGS. 16A and 16B are schematic diagrams illustrating an example of a method of forming a silicide layer under a semiconductor pillar pattern.

Referring to FIG. 16A, in a state in which the first groove G1 is formed, a metal material 155 may be formed along the surface of the first groove G1. The metal material 155 may be formed by an ALD (atomic layer deposition) method or

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the like, and may be formed to a thickness that does not completely fill the first groove G1.

Subsequently, when a heat treatment is performed, the metal material **155** may react with the semiconductor material of the semiconductor pillar **120A** to form the silicide layer **156**.

Referring to FIG. **16B**, the metal material **155**, which has not been reacted, may be removed, and subsequent processes, for example, the process of FIGS. **5A** and **5B** and subsequent processes thereof, may be performed.

In the present embodiment, the silicide layer **156** may be formed after the formation of the first groove G1, but the present disclosure is not limited thereto. In another embodiment, a silicide layer may be formed after the second groove G2 is formed (refer to FIGS. **6A** and **6B**).

Although the above embodiments have described the cell region in which the transistor and the storage node **240** connected thereto are arranged, the semiconductor device may further include a peripheral circuit region for driving the cell region or performing other functions. This peripheral circuit region may be formed together with the cell region. This will be described with reference to FIGS. **17A** to **17F**.

FIGS. **17A** to **17F** are schematic diagrams illustrating a semiconductor device according to another embodiment of the present disclosure, and a method for fabricating the same. FIGS. **17A** to **17F** are cross-sectional schematic diagrams taken along the aforementioned line C-C'. Parts substantially identical to those of the above-described embodiments are denoted by the same reference numerals.

Referring to FIG. **17A**, after providing a substrate **100** having a cell region CA and a peripheral circuit region PA, the above-described processes of FIGS. **1A** to **3D** may be performed to provide a stacked structure of a sacrificial layer **110**, a semiconductor pillar **120A**, and a hard mask layer **140** over the substrate **100** in the cell region CA. At this time, substantially the same processes may be performed in the peripheral circuit region PA, so that a peripheral sacrificial layer **1100**, a peripheral semiconductor layer **1200A**, and a peripheral hard mask layer **1400** may be formed over the substrate **100** in the peripheral circuit region PA. Accordingly, the peripheral sacrificial layer **1100**, the peripheral semiconductor layer **1200A**, and the peripheral hard mask layer **1400** may be positioned in the same level in the vertical direction and be formed of the same material as the sacrificial layer **110**, the semiconductor pillar **120A**, and the hard mask layer **140**, respectively. However, the arrangement or shape of the peripheral sacrificial layer **1100**, the peripheral semiconductor layer **1200A**, and the peripheral hard mask layer **1400** in a plan view, may be variously modified depending on the configuration of a device/circuit to be formed in the peripheral circuit region PA.

Referring to FIG. **17B**, in the process of FIGS. **4A** to **4D**, for example, in the process of removing the sacrificial layer **110** of the cell region CA to form the first groove G1, the peripheral sacrificial layer **1100** may also be removed.

Referring to FIG. **17C**, an initial second insulating layer **150'** may be formed over the substrate **100** of the cell region CA and the peripheral circuit region PA to fill the first groove G1 and a space formed by the removal of the peripheral sacrificial layer **1100** while filling an empty space thereover.

The initial second insulating layer **150** may be formed by depositing an insulating material over the structure of FIG. **17B**, and performing a planarization process until the surfaces of the hard mask layer **140** and the peripheral hard mask layer **1400** are exposed. For reference, when a portion of the initial second insulating layer **150'** in the cell region

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CA is removed by a subsequent process, it may be transformed into the second insulating layer **150** illustrated in FIGS. **5A** to **5D**.

Referring to FIG. **17D**, a first mask pattern M1 covering the peripheral circuit region PA and exposing the cell region CA may be formed over the structure of FIG. **17C**, and aforementioned processes described with reference to FIGS. **5A** to **12D** may be performed. Accordingly, the same structure as the structure shown in FIGS. **12A** to **12D** may be secured in the cell region CA.

Since the peripheral circuit region PA is in a covered state, the state of FIG. **17C** may be maintained.

Referring to FIG. **17E**, a second mask pattern M2 may be formed over the structure of FIG. **17D**. The second mask pattern M2 may cover the cell region CA and may leave the peripheral circuit region PA uncovered or exposed. Then, a planarization process may be performed so that the peripheral semiconductor layer **1200A** of FIG. **17D** is exposed. The planarization process may be performed until the peripheral semiconductor layer **1200A** has a desired thickness on the premise that the peripheral hard mask layer **1400** is removed. The planarized peripheral semiconductor layer **1200A** will be referred to as a peripheral semiconductor layer pattern **1200B**.

The thickness of the peripheral semiconductor layer pattern **1200B** may be less than or equal to the thickness of the peripheral semiconductor layer **1200A**. The planarization process may include a thinning process for making the thickness of the peripheral semiconductor layer pattern **1200B** to be, for example, less than 10 nm. In the vertical direction, the bottom surface of the peripheral semiconductor layer pattern **1200B** may be positioned at the same or substantially the same level as the bottom surface of the semiconductor pillar pattern **120B**, and the top surface of the peripheral semiconductor layer pattern **1200B** may be positioned at the same or substantially the same level as the top surface of the semiconductor pillar pattern **120B** or at a lower level than the top surface of the semiconductor pillar pattern **120B**. The peripheral semiconductor layer pattern **1200B** may function as an active region of the peripheral circuit region PA.

During the planarization process on the peripheral semiconductor layer **1200A**, the initial second insulating layer **150'** may also be planarized to have the same or substantially the same top surface height as the peripheral semiconductor layer **1200A**. The initial second insulating layer **150'** may surround side and bottom surfaces of the peripheral semiconductor layer pattern **1200B**. The initial second insulating layer **150'** may have a bottom surface height substantially equal to that of the second insulating layer pattern **150A** and a higher top surface height than that of the second insulating layer pattern **150A**.

Referring to FIG. **17F**, a desired device may be formed using the peripheral semiconductor layer pattern **1200B**. In the present embodiment, a case in which a flat transistor is formed in the peripheral circuit region PA is illustrated. As an example, a peripheral circuit gate insulating layer **1500** may be formed along the bottom profile on the structure of FIG. **17E**, and a peripheral circuit gate pattern **1600** and a peripheral circuit gate spacer **1700** may be formed thereover.

According to the present embodiment, since the peripheral semiconductor layer pattern **1200B** functioning as an active region of the peripheral circuit region PA is physically and electrically separated from the substrate **100** by the initial second insulating layer **150**, it may be possible to form a fully depleted silicon on insulator (SOI) transistor. In this case, it may be advantageous in terms of variation in

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operating characteristics of the transistor and reduction in leakage. When the thickness of the peripheral semiconductor layer pattern 1200B is, for example, less than 10 nm, the fully depletion characteristic may be further improved.

The above-described embodiments may be applicable to any semiconductor device including an NMOS transistor, a PMOS transistor, or a CMOS transistor, and a method for fabricating the same. The above-described embodiments may be applicable to various semiconductor devices, for example, non-volatile memory such as flash memory, resistive random-access memory (RRAM), phase-change random-access memory (PRAM), or magneto resistive random-access memory (MRAM), volatile memory such as dynamic random-access memory (DRAM) or static random-access memory (SRAM), non-memory such as a logic circuit, a CMOS image sensor (CIS), or the like.

According to the above embodiments of the present disclosure, it may be possible to provide a semiconductor device capable of increasing the degree of integration and improving operating characteristics, and a method for fabricating the same.

Although various embodiments have been described for illustrative purposes, it will be apparent to those skilled in the art that various changes and modifications may be made without departing from the spirit and scope of the present teachings as defined in the following claims.

What is claimed is:

1. A semiconductor device comprising:
 - a substrate;
 - a plurality of semiconductor pillars disposed over the substrate and arranged in a first direction and a second direction crossing the first direction;
 - an insulating layer pattern disposed between the substrate and the semiconductor pillars and extending in the second direction;
 - a first conductive line disposed between the insulating layer pattern and the semiconductor pillars and extending in the second direction;
 - a second conductive line formed over sidewalls of the semiconductor pillars and extending in the first direction; and
 - a storage node disposed over each of the semiconductor pillars.
2. The semiconductor device according to claim 1, wherein the first conductive line is physically and electrically separated from the substrate by the insulating layer pattern.
3. The semiconductor device according to claim 1, wherein both sidewalls of the first conductive line are aligned with both sidewalls of the insulating layer pattern in the first direction.
4. The semiconductor device according to claim 1, wherein the semiconductor pillars have a different material or a different crystal structure from that of the substrate.
5. The semiconductor device according to claim 1, further comprising:
 - a conductive pattern interposed between each of the semiconductor pillars and the storage node,

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wherein an area of a top surface of the conductive pattern is larger than an area of a top surface of each of the semiconductor pillars in a plan schematic diagram.

6. The semiconductor device according to claim 1, further comprising:
 - a junction region formed at a bottom end of each of the semiconductor pillars, which is in contact with the first conductive line.
7. The semiconductor device according to claim 1, further comprising:
 - a silicide layer formed at a bottom end of each of the semiconductor pillars, which is in contact with the first conductive line.
8. The semiconductor device according to claim 1, wherein the substrate includes a cell region and a peripheral circuit region, and
 - the semiconductor pillars, the insulating layer pattern, the first conductive line, the second conductive line, and the storage node are disposed in the cell region, and the semiconductor device further comprising:
 - a peripheral semiconductor layer pattern disposed in the peripheral circuit region over the substrate; and
 - an insulating layer disposed in the peripheral circuit region over the substrate and surrounding side and lower surfaces of the peripheral semiconductor layer pattern.
9. The semiconductor device according to claim 8, wherein the bottom surface of the peripheral semiconductor layer pattern is located at the same height as a bottom surface of each of the semiconductor pillars, and
 - a top surface of the peripheral semiconductor layer pattern is located at a height equal to or lower than a top surface of each of the semiconductor pillars.
10. The semiconductor device according to claim 8, wherein the semiconductor pillars and the peripheral semiconductor layer pattern are formed of the same material.
11. The semiconductor device according to claim 8, wherein the peripheral semiconductor layer pattern is physically and electrically separated from the substrate by the insulating layer.
12. The semiconductor device according to claim 8, wherein a bottom surface of the insulating layer is located at the same height as a bottom surface of the insulating layer pattern, and
 - a top surface of the insulating layer is located at a height higher than a top surface of the insulating layer pattern.
13. The semiconductor device according to claim 8, wherein the insulating layer and the insulating layer pattern are formed of the same material.
14. The semiconductor device according to claim 8, further comprising:
 - a peripheral circuit gate insulating layer formed over the peripheral semiconductor layer pattern; and
 - a peripheral circuit gate pattern formed over the peripheral circuit gate insulating layer.

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